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BANK OF THAILAND

MONETARY POLICY REPORT

March 2018

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
March 2018

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Porametee Vimolsiri	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2018. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

Monetary Policy Conduct in the First Quarter of 2018

The Committee assessed that the Thai economy continued to gain further traction, driven by improved growth in exports and tourism in tandem with the global economy and domestic demand. In addition, private consumption would continue to gradually increase. Headline inflation would trend up albeit at a slightly slower pace than previously assessed due to large supply of meat, vegetables, and fruits in the market which led to a decline in prices. The increase in the buildup of risks to financial stability from a prolonged low interest rate environment in certain pockets still warranted close monitoring. Against this backdrop, **the Committee had to weigh various factors in formulating the most appropriate course of monetary policy in order to attain the objectives of sustainable economic growth and price stability while preserving financial stability. The Committee voted unanimously and later 6 to 1 to keep the policy rate unchanged at 1.50 percent at the meetings on February 14 and March 28, 2018 respectively.** In deliberating their decision, the Committee judged that accommodative monetary policy stance would still be necessary to support a more robust growth in domestic demand which would foster the return of headline inflation to target. Moreover, the Committee viewed that the policy rate at 1.50 percent remained appropriate in that it would facilitate sufficiently accommodative financial conditions, as reflected in low real interest rates and continued improvements in private sector financing through commercial banks and financial markets. One committee voted to raise the policy rate by 0.25 percentage point to 1.75 percent at the meeting on March 28, 2018 as the unusually low interest rates for a prolonged period might prompt households and businesses to underestimate risks to potential changes in financial conditions in the future. Furthermore, a gradual, timely reduction in monetary policy accommodation would not hinder economic growth or price stability. It would instead help alleviate risks to financial stability in the future and facilitate a smooth process of monetary policy normalization.

Looking ahead, the Committee deemed that monetary policy accommodation should be maintained for some time to support a more robust economic growth and foster the gradual return of headline inflation toward the medium-term target. The Committee would stand ready to utilize available policy tools to facilitate the return of headline inflation to target in an appropriate time, while seeking to achieve sustainable economic growth and maintaining financial stability.

Assessments of the Economic and Inflation Outlooks as the Basis for Policy Formulation

1. Global Economy

The global economy was projected to continue expanding and remained a driver of Thai exports in the period ahead. Advanced economies would expand further due mainly to consumption and exports. Moreover, the U.S. would exhibit a higher growth than previously expected on account of the U.S. tax reform policy, a strong labor market, and improved consumer confidence. The euro area would also record a stronger growth than the previous assessment on the back of exports and domestic demand. In addition, accommodative financial conditions, improved private sector confidence, and a gradual recovery in the labor market conditions would continue to support growth going forward. Japan would see growth across all economic sectors with private consumption expected to rise from robust consumer confidence and improved labor market conditions. The Chinese economy was expected to gain further traction despite some slowdown, consistent with its economic growth target for 2018 and the government's economic reforms. Meanwhile, other Asian economies would continue expanding following robust growth in exports across various product categories which would underpin employment and household consumption in the period ahead. **The Committee therefore revised up the growth forecast for Thailand's trading partners from 3.5 percent to 3.7 percent in 2018.** However, there remained certain risks that warranted monitoring including (1) an increase in import tariffs by the U.S. which might lead to retaliatory measures by major economies which could cause a direct impact on trade volume and an indirect impact on prices as well as productions in global supply chains, (2) uncertainties surrounding other U.S. economic policies such as infrastructure investment policy, and (3) geopolitical risks that could intensify and undermine the economy and financial markets.

Most central banks maintained accommodative monetary policy stance. Nonetheless, some central banks gradually reduced the degree of monetary policy accommodation following improvements in economic and inflation outlooks. The Federal Reserve (Fed) raised its policy rate in March 2018 as the market expected, while the European Central Bank (ECB) adjusted its forward guidance, reflecting the possibility of a sooner-than-expected reduction in the degree of monetary policy accommodation in the future. Moreover, a number of central banks in Asia started to signal a monetary policy normalization in the period ahead given improved economic growth and a slow rise in inflation.

2. Financial Conditions and Financial Stability

Financial conditions remained accommodative. The market correction which resulted in lower asset prices in the global financial market did not have a significant effect on Thailand's financial markets. Short-term bond yields remained below the policy rate due to a decreased supply of short-term bonds and an increased demand from foreign investors. Medium- and long-term bond yields declined at the beginning of the year owing to higher demand from both Thai and foreign investors. Subsequently, the yields gradually edged up in line with U.S. government bond yields, albeit remaining low due to following reasons. First, net supply of Thai government bonds was low as a number of bonds reached their maturities during the first quarter of the year. Second, Thai government bond yields were more attractive relative to other regional countries and thus continued to attract foreign investors into Thai bond markets. Third, Thailand's inflation expectations remained low. Meanwhile, interest rates on new loans (NLR) remained at low levels, reflecting accommodative financial conditions. Overall private credit expanded on the back of both corporate and household sectors. In particular, SME loan growth improved across various sectors, reflecting a more broad-based economic recovery. Furthermore, business financing through bond and equity markets continued to increase. Meanwhile, household loans expanded in all categories, particularly auto leasing. With regard to exchange rates, the baht appreciated against the U.S. dollar from the previous quarter due to the weakening of the U.S. dollar. Such appreciation was in line with movements of most regional currencies, but the pace of appreciation varied depending on country-specific factors. The real effective exchange rate (REER) somewhat appreciated consistent with improving economic fundamentals.

Financial stability remained sound but there remained pockets of risks that warranted monitoring. These included, first, some sign of accelerating household debt while debt serviceability of some households continued to deteriorate. Second, some SMEs which continued to incur losses would see deterioration in their abilities to service debt and cushion against economic volatilities. Third, the search-for-yield behavior persisted given a prolonged low interest rate environment which could lead to underpricing of risks. Such behaviour included, for instance, a large expansion of foreign investment funds (FIF) with high concentration in some countries. Moreover, assets and deposits surged among savings cooperatives despite some slowdown after the revised regulations to upgrade supervision practices. Fourth, there was also an oversupply of property in certain price ranges and areas. This was particularly the case for condominiums with price below three million baht whose supply continued to rise.

3. Economic and inflation outlook

The Thai economy was projected to continue expanding and record 4.1 percent growth in 2018 and 2019, higher than previously estimated in the previous Monetary Policy Report. The upward revision was on account of a continued improvement in merchandise exports and tourism, thanks to robust growth of trading partners and a gradual expansion in private spending, which would be partly supported by government measures aimed at improving the well-being of low-income households.

Merchandise exports continued to expand across various product categories and almost all export destinations. The value of merchandise exports in 2018 was revised to grow by 7.0 percent, up from 4.0 percent assessed in the previous quarter, in tandem with economic expansion of trading partners. Such improvement was observed particularly among exports of electronics, auto parts, and processed agricultural products. Electronics exports would see growth especially in electronic components related to the Internet of Things (IoT) and electronic parts used in automobile. Nonetheless, the increase in import tariffs by the U.S. was expected to have a limited impact on Thai exports in the short term as only a small share of Thai products in total Thai exports would be subject to these new tariffs. However, downside risks to export growth increased as retaliatory measures could impact major trading partners' growth, especially through supply chains, thereby weighing on international trade and Thai exports in the period ahead. Moreover, export prices, especially for commodities and oil-related products,

would likely trend up in line with oil prices. Meanwhile, import value would also rise following increased demand for raw materials and intermediate goods consistent with the improved growth and higher oil prices.

Exports of services in 2018 were expected to expand in line with continued improvement in tourism. The projection of the number of foreign tourists was revised up to 37.6 million in 2018, up from the previous forecast of 37.3 million. The upgrade was on account of a steady growth in the number of Chinese tourists, both group and independent tourists. In addition, Indian tourists would likely increase in the period ahead after the Indian authority granted Thai airlines additional capacity entitlements under bilateral air services agreements since March 2018. However, Thailand still faced capacity constraints in accommodating the rising number of tourists in the main airport, although such constraints were somewhat lessened in recent periods with increasing roles of local airports operated by the Department of Airports and the Ministry of Defense, especially Krabi, Suratthani, and U-tapao Airports, in facilitating international flights.

Private consumption would gradually expand in the period ahead, supported by income among non-agricultural households with medium-to-high income. In addition, signs of improvement in earnings of low-income households began to be observed as employment picked up in export-related businesses and the tourism sector. On the other hand, household income in some agricultural sectors was projected to slow down in line with agricultural prices. Nevertheless, overall household purchasing power gradually improved. This was partly a result of structural changes in the labor market such as adoption of automation in place of human labor in the production process as well as a decline in agricultural prices which made farming unattractive. Hence, these led to greater migration of workers out of the manufacturing and agricultural sectors to the services sector. Nevertheless, some workers might earn less as they migrated to certain parts of the services sector with relatively lower wages. Moreover, elevated household debt underlined the fact that workers still needed to earmark part of their income for debt repayment. Nonetheless, government measures aimed at improving the well-being of the low-income households—such as the social welfare card project (second phase) and the community enterprise development project—would partly help raise private consumption in the period ahead.

Public spending remained a driver of economic growth on the back of expansion in public consumption and investment. However, public expenditure was lower than previously estimated because the impact from the Public Procurement and Supplies Management Act, B.E. 2560 on government disbursement was larger and longer than expected. In particular, disbursement by certain state agencies, especially local governments, was significantly affected, where those agencies would need some times to adjust. Meanwhile, most investment projects of state-owned enterprises (SOEs) were not affected by the new law, but some projects were postponed due to delays in contract signing and problems in accessing construction sites.

Private investment would continue to improve, consistent with growth of exports and private consumption. Such improvement was partly reflected by a rising demand for business credits and accelerated value of investment application submitted to the Board of Investment at the end of 2017. Moreover, investment incentives offered by the government would play an important role in driving private investment, particularly through public-private partnership, government infrastructure investment projects, and the Eastern Economic Corridor (EEC) initiative whose prospects became more certain. Therefore, this would help promote business confidence and also attract foreign investment.

Headline inflation remained low due to supply-side factors but was expected to slowly rise. In recent periods, headline inflation stabilized at a lower level than previously estimated. This was owing to a contraction in fresh food prices following declined prices of meat, vegetables and fruits, stemming from a large supply in the market. Going forward, headline inflation would slowly rise as cost-push inflationary pressures increased at a slower pace given that fresh food prices were projected to remain low, while the impact from minimum wages increases in 2018 on inflation was largely limited. Moreover, demand-pull inflationary pressures remained close to the previous estimate. **The Committee therefore projected headline inflation to average 1.0 and 1.2 percent in 2018 and 2019 respectively, while core inflation was projected to average 0.7 and 0.8 percent in 2018 and 2019 respectively.**

Risks to the growth projection were expected to tilt to the downside. The Thai economy would face greater risks from the U.S. foreign trade policy and retaliatory measures from major economies. As a result, this might

weigh on trading partners' growth to a larger extent than previously assessed, despite some improvements in geopolitical risks related to the Korean Peninsula. Moreover, domestic risks included a larger-than-expected impact from the Public Procurement and Supplies Management Act on public expenditure, and a lower-than-expected domestic spending as improvement in purchasing power was not yet sufficiently broad-based. However, on the upside, there were possibilities that the growth outturn might be higher than the baseline projection given a better-than-expected growth outlook of Thailand's trading partners with supporting factors including the U.S. tax reform, Chinese government measures which could soften its economic slowdown, better-than-expected Asian exports, as well as government infrastructure investment which could stimulate a higher private investment than the baseline projection. **Risks to the inflation projection were also assessed to tilt to the downside** consistent with risks to the growth forecast.

1. Global Economy

Advanced economies continued to expand driven primarily by consumption and exports. Meanwhile, risks to global economic growth tilted to the downside as the U.S. foreign trade protectionist policy could intensify as well as likelihood and impact of retaliation by trading partners could be greater than expected.

The U.S. economy was expected to record a stronger growth than the previous assessment on the back of the tax reform policy which would support private spending. Consumption continued expanding thanks to a strong labor market and improved consumer confidence. In addition, sound household financial positions and the reduction of individual income tax would underpin consumption going forward. At the same time, private investment continued to expand in line with robust business confidence, steady improvements in corporate profits, and the expansion in production sectors. Moreover, private investment would further benefit from the lower corporate income tax and repatriation tax. **Euro area economies were projected to exhibit a higher growth than the previous estimate and started to be more broad-based** thanks to the expansion of exports in tandem with global trade as well as continued domestic demand growth. Moreover, accommodative financial conditions, improved private sector confidence, and a gradual recovery in the labor markets would help support economic growth in the period ahead. **Japan continued to see growth across all economic sectors.** Private consumption would continue expanding thanks to robust consumer confidence and improved labor market conditions. In addition, exports were expected to rise in accordance with steady improvements in global trade.

Looking ahead, growth of advanced economies would continue to face risks from (1) uncertainties surrounding U.S. foreign trade policy, where an increase in import tariffs on solar cells, large washing machines, and metals and aluminum could lead to retaliatory measures by major economies, despite the limited impact on the economy, and would in turn undermine global trade volume, prices, and supply chain, as the U.S. recently announced to raise tariff on imported goods from China which led to retaliation, (2) uncertainties pertaining to other policies in the U.S. such as infrastructure investment policy, and (3) negotiations on trade and other agreements between the UK and the European Union after Brexit.

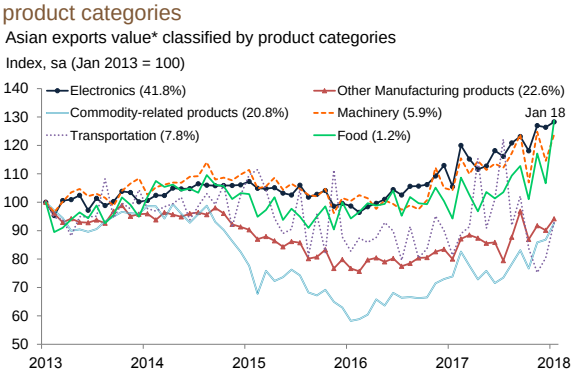
The Chinese economy was expected to gain further traction despite its potential economic slowdown. Meanwhile, other Asian economies would continue expanding following robust exports growth in tandem with global demand and a gradual recovery of domestic demand thanks to improved private sector confidence.

In the fourth quarter of 2017, China's growth remained close to the previous quarter, driven mainly by exports and consumption. In the period ahead, however, the Chinese economy was expected to slow down, consistent with its economic growth target for 2018 and the government's economic reforms. Such reforms particularly included (1) economic structural reforms that placed emphasis on quality growth through reducing excess production capacity and shutting down environmentally unfriendly factories and (2) financial stability measures such as measures to curb speculation in the real estate sector, control shadow banking risks, as well as tighten credit standards on loans given to local government,

as these loans were partly contributed to high levels of corporate debt in the previous period especially those owned by state-owned enterprises.

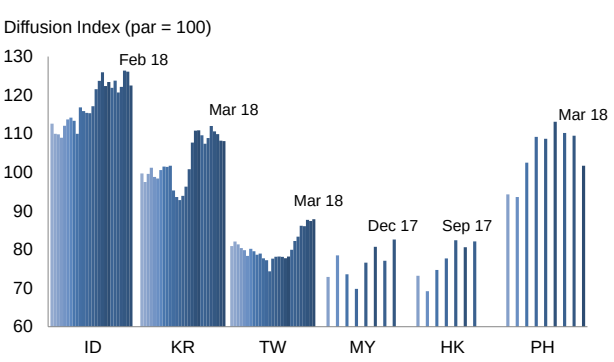
Asian economies (excluding Japan and China) continued to expand driven mainly by higher-than-expected exports growth across various product categories, particularly electronics, machinery and equipment, and commodities and food (Chart 1.1). Such export expansion and the broader growth helped support manufacturing to gradually recover and boost corporate performance and confidence, which would help underpin employment. In addition, fiscal impetus also supported household consumption going forward. Looking ahead, Asian economies were expected to gain further traction although there remained several risks. First, uncertainties surrounding U.S. foreign trade policy could lead to retaliatory measures between the U.S and major economies despite the limited direct impact from increased U.S. import tariffs due to a small share of those Asian exports to the U.S. However, a potential retaliation could adversely affect global trade volume and cause an indirect impact on Asian exports. Second, elevated household debt could weigh on domestic demand recovery in several countries. Third, China's economic reforms could slow down demand from China more than expected which would affect overall Asian exports. Fourth, geopolitical risks in the Middle East and the Korean Peninsula would continue to warrant monitoring despite some improvement.

Chart 1.1 Asian exports continued to expand across various product categories



Note: * Asian exports include HK, TW, KR, MY and SG.
( ) a share of total exports in 2017
Commodity-related products include crude oil, metals, chemicals, rubber, and vegetable oil.
Other manufacturing products include textile, papers, furniture, footwear and miscellaneous
Source: CEIC

Chart 1.2 Asian consumer confidence continued to improve



Note: Philippines' consumer confidence is scaled from par = 0 to par = 100
Source: CEIC and calculation by the Bank of Thailand

The growth outlook of Thailand's trading partners was revised up, but risks to growth projection were expected to tilt to the downside.

Economic growth of Thailand's trading partners would likely to be stronger than expected in the previous *Monetary Policy Report*. This was due to better-than-expected economic outturns in the fourth quarter of 2017 in many countries, coupled with continued exports growth and U.S. fiscal policy. The Committee therefore revised up the growth forecast for Thailand's trading partners to 3.7 percent in 2018 and projected the growth forecast for 2019 at 3.6 percent (Table 1.1).

The Committee assessed that risks to growth of Thailand's trading partners would tilt to the downside owing to U.S. foreign trade policy that could lead to retaliatory measures by major economies. In addition, there remained risks that warranted monitoring including (1) uncertainties surrounding other U.S. economic policies and (2) geopolitical risks that could

intensify and lead to increased volatility in financial markets, commodity prices, and the real sector, despite recent easing in tensions in some areas particularly the Korean Peninsula.

Table 1.1 Assumption on trading partners' economic growth

Annual change (%YoY)	Weight (%)	2017*	2018	2019
United States	14.9	2.3	2.7 (2.5)	2.4
Euro area	10.0	2.5	2.3 (1.8)	2.0
Japan	13.6	1.7	1.5 (1.2)	1.0
China	15.7	6.9	6.6 (6.5)	6.4
Asia (excluding Japan and China)**	37.4	4.6	4.2 (4.1)	4.3
Total***	100	3.9	3.7 (3.5)	3.6

Note: *Outturn

**Weighted by a share of Thailand's total exports to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)

***Weighted by a share of Thailand's total exports to major 13 trading partners as of 2014 (including the United Kingdom and Australia)

( ) reported in Monetary Policy Report December 2017

Most central banks maintained accommodative monetary policy stance. However, some central banks gradually reduced the degree of monetary policy accommodation following improvements in growth and inflation outlooks.

Most central banks maintained accommodative monetary policy stance. However, some central banks gradually reduced the degree of monetary policy accommodation. For instance, Bank Negara Malaysia (BNM) raised its policy interest rate in January 2018 due to more robust economic growth which was driven mainly by domestic demand. The Federal Reserve (Fed) raised the federal funds rate at its latest meeting in March 2018 in line with market expectations. The European Central Bank (ECB) adjusted its forward guidance by removing the easing bias on the asset purchase program, pointing to a possibility that the ECB could reduce the degree of monetary policy accommodation sooner than market expectations. Moreover, several central banks of some Asian economies started to signal monetary policy normalization in the period ahead owing to a continuation of economic growth and a gradual increase in inflation.

Capital inflows toward emerging markets (EMs) accelerated in early 2018 as foreign investors allocated a larger share to EMs assets. However, capital flows subsequently recorded net outflows due to market corrections and uncertainties pertaining to U.S. foreign trade policy.

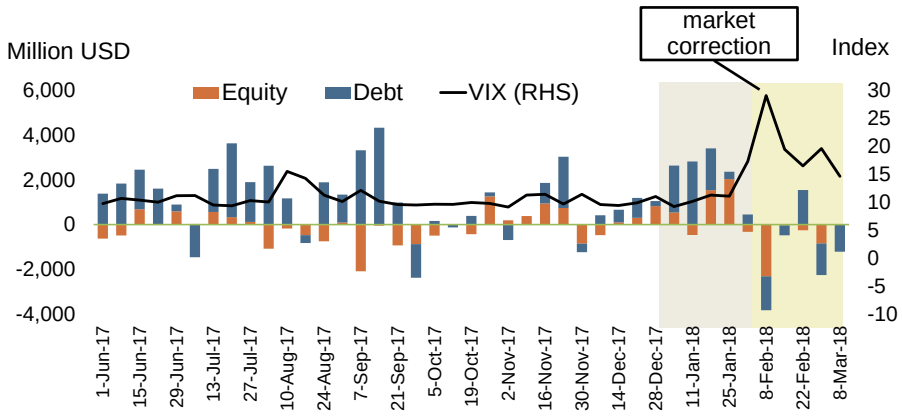
In January 2018, capital inflows to emerging markets (EMs) accelerated relative to the end of last year as a result of portfolio rebalancing of foreign investors with a larger share of EMs assets at the beginning of the year. This was on the back of improved growth outlooks in EMs, especially in Asia. Nevertheless, in February 2018, EM capital flows recorded net outflows largely due to market correction which resulted in a drop in worldwide asset prices

and a short-term surge in the volatility index (VIX)[1/] (Chart 1.3). In particular, the recent market corrections occurred after U.S. economic indicators including employment and inflation turned out to be better than expected. Consequently, investors anticipated that the Fed would raise the federal funds rate sooner than previously assessed. This was reflected in a higher 10-year U.S. Treasury yield which prompted investors to sell risky assets globally. Later in March 2018, net capital outflows continued owing to uncertainties surrounding U.S. foreign trade policy, particularly the announcement to impose import tariffs on metal and aluminum products. In addition, reduction in the degree of monetary policy accommodation by several central banks was expected to tighten liquidity in the global financial markets going forward. Hence, foreign investors became more cautious about their investment decisions and especially took into account country specific factors such as economic fundamentals and external stability.

Looking ahead, global financial markets would likely remain volatile. International capital flows could move both into and out of Thailand given uncertainties on the external front including U.S. foreign trade policy and retaliation by major countries, monetary policy directions of major advanced economies, and geopolitical risks.

Chart 1.3 Capital inflows to EMs surged during beginning of the year due to portfolio rebalancing of foreign investors; however, market correction and uncertainties pertaining to U.S foreign trade policy led to capital outflows from EMs since February

Capital inflows to EMs* (weekly)



Note: *EMs includes Thailand, Indonesia, India, South Africa, and Turkey

Sources: Bloomberg and Institute of International Finance

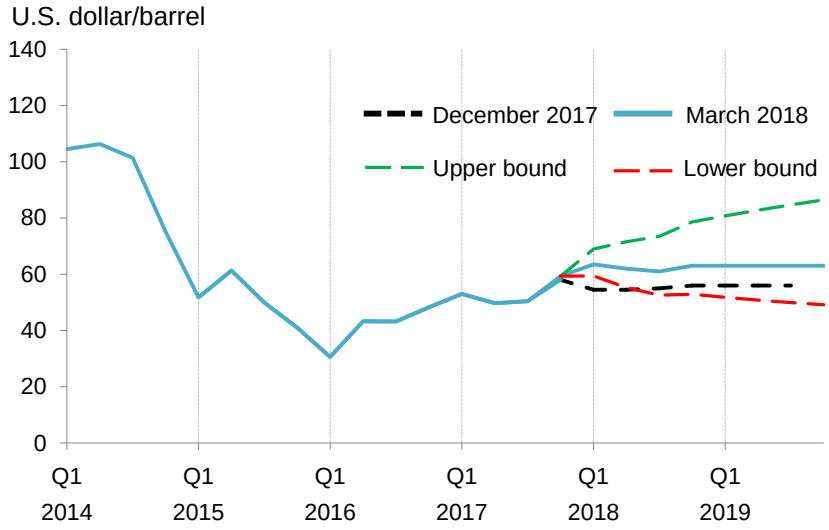
[1/] VIX measures volatility in stock prices and is calculated from implied volatility of options in the S&P 500.

Crude oil prices trended up due to a decline in crude oil stock in the first quarter of 2018 but oil prices would likely decline during the remaining of the year as a result of higher shale oil production. In 2019, crude prices were projected to stabilize at a high level compared with the previous three years due to rising demand for oil following global economic growth.

The Dubai crude prices steadily increased in the first quarter of 2018. Crude oil stock ran down toward the equilibrium level as oil demand rose in line with global economic growth. Moreover, compliance with the production cuts agreement among OPEC and non-OPEC producers would limit an increase in oil supply. **However, the Committee assessed crude oil prices to gradually decline in the remaining of 2018** as U.S. shale oil producers would increase production following higher prices in the previous period. **In 2019, oil prices were expected to stabilize at a high level compared with the previous three years** in accordance with rising demand for oil following global economic growth. Meanwhile, higher U.S. shale oil production continued to put downward pressures on crude oil prices.

The Committee therefore revised up the projection for Dubai crude oil prices from 55.0 to 62.4 U.S. dollars per barrel in 2018 while prices were forecast to average at 63.0 U.S. dollars per barrel in 2019. The Committee viewed that risks to the projection still tilted to the upside owing to higher-than-expected oil demand in tandem with global economic growth and geopolitical risks in the Middle East and the Korean Peninsula that could raise oil prices. However, some downside risks remained due to higher-than-expected supply from U.S. shale oil producers and higher production by OPEC and non-OPEC producers toward the end of the production cut agreement. These factors might lead oil prices to be lower or rise more slowly than the baseline projection.

Chart 1.4 Dubai crude oil price in 2018Q1 continued to increase from previous quarter as excess oil supply declined toward equilibrium level



2. Thai Economy

2.1 Recent Developments

The Thai economy continued to gain further traction driven by external demand and gradual improvements in domestic demand.

The Thai economy expanded 4.0 percent in the **fourth quarter of 2017**, driven mainly by expansion of merchandise exports across all major export destinations and almost all product categories on account of sustained improvements in external demand. Exports of services also recorded strong growth on the back of increased foreign tourists in almost all groups, particularly Chinese and Russian tourists. Nonetheless, the number of Malaysian tourists continued to contract, partly as a result of intensifying competition to attract tourists from Vietnam. Private consumption continued expanding, especially spending on durable goods and was supported by strong purchasing power of non-agricultural households, especially those in the services sector, and by government measures such as the social welfare card project. However, overall purchasing power was not robust as benefits from economic expansion had yet to fully extend to all households. Private investment improved from investment in machinery and equipment, although investment in the construction sector continued to contract. Public expenditure declined from a lower public investment, partly owing to limited disbursement efficiency of some government agencies and a longer procurement process as a result of the Public Procurement and Supplies Management Act, B.E. 2560. Overall, the Thai economy recorded a 0.5 percent growth from the previous quarter, after seasonal adjustment, in the fourth quarter of 2017 and expanded 3.9 percent in 2017, an improvement from 3.3 percent in 2016.

The Thai economy continued to gain further traction in the **first quarter of 2018**, as reflected in recent economic indicators. In particular, growth of merchandise exports was robust and broad-based across all sizes of exporters. The tourism sector continued to expand on account of an increasing number of foreign tourists in almost all groups consistent with the stronger global economic growth. Private consumption continued to improve, especially spending on services and durable goods, thanks to spending of non-agricultural households with high income. On the other hand, household income in some agricultural sectors declined, mainly due to a fall in agricultural prices despite increased agricultural output. Meanwhile, manufacturing, particularly in automobiles, grew in line with continued expansion in demand. Private investment picked up in tandem with economic expansion, particularly investment in machinery and equipment, while there were signs of improvements in investment in the construction sector. Public expenditure resumed some growth on account of investment expenditure by the main government agencies including the Department of Highways, the Department of Rural Roads, and the Department of Irrigation. However, current expenditure contracted slightly due to decreased expenditure on goods and services by the Office of the Basic Education Commission.

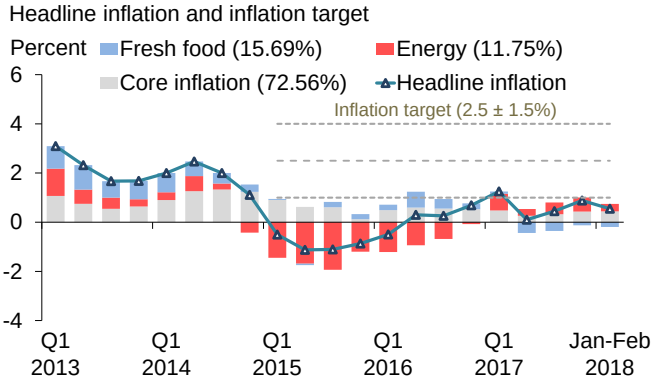
Benefits from economic expansion were increasingly distributed to the labor market. Earnings of non-agricultural low-income households began to show signs of improvement, while household income in some agricultural sectors declined given that prices of several agricultural products remained subdued. Overall purchasing power improved slowly.

Non-agricultural household income improved overall. Purchasing power of the high-income group continued to increase, while purchasing power of the low-income group in certain export-related businesses, such as food and rubber products, began to improve. However, there were structural changes in the labor market with increased migration of non-farm workers out of the manufacturing sector to the services sector, particularly tourism-related businesses. Thus, some workers might earn less as they migrated to the services sector where wages were mostly lower than in the manufacturing sector. Income of some agricultural households declined as prices of several agricultural products, such as rubber, palm oil, and livestock, remained subdued. Nonetheless, prices of rice improved, both white and jasmine rice, on the back of increased external demand. As a result, overall household purchasing power was not sufficiently strong. With regard to the progress in managing and legalizing immigrant workers in Thailand in accordance with the Royal Decree on Managing the Work of Aliens, the government announced the extension of the nationality verification period to June 30, 2018. This partly helped lessen adverse impacts on the economy, as most businesses were able to adjust and carry on their operations.

Headline inflation declined due to lower energy and fresh food prices, while core inflation edged up slightly due to the excise tax increase and a gradual recovery in domestic demand.

Headline inflation averaged 0.56 percent over the first two months of 2018. A continued contraction in fresh food prices was the main factor keeping headline inflation at a low level. Nevertheless, an average headline inflation over the first two months was lower than an average of 0.88 percent recorded in the fourth quarter of 2017 (Chart 2.1). This was attributable to a slowdown in energy price increases in line with lower global crude prices and a contraction in fresh food prices as prices of meat and fruit declined given a large output in the market.

Chart 2.1 Headline inflation declined slightly from previous quarter due mainly to energy and fresh food prices

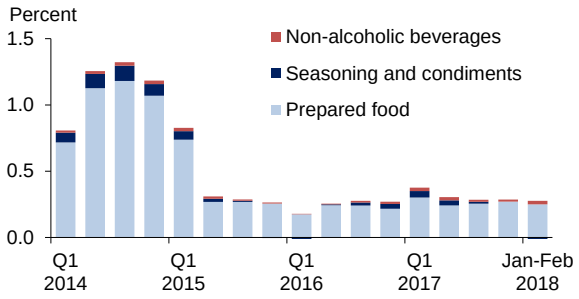


Note: ( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculation by Bank of Thailand

Core inflation over the first two months of the first quarter averaged 0.61 percent, largely unchanged from the previous quarter. Prices of food items in the core inflation basket stabilized at a low level (Chart 2.2) in line with low costs of fresh food and liquefied petroleum gas (LPG). Meanwhile, prices of non-food components in core inflation edged up slightly (Chart 2.3) following an increase in the excise tax on tobacco and alcoholic beverages. Prices of other items slowly expanded in tandem with a gradual improvement in domestic demand (Chart 2.4). In addition, structural factors, including production technology developments, e-commerce, and globalization, led to lower costs of production and intensified price competition. These factors weighed on an increase in prices of non-food components in core inflation in recent periods. Short-term (one-year ahead) inflation expectations according to a survey of businesses in February 2018 stood at 2.1 percent, largely unchanged from the fourth quarter of 2017. Meanwhile, long-term (five-year ahead) inflation expectations according to a

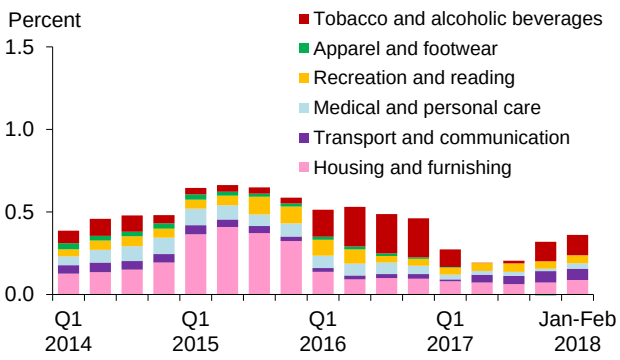
survey of professional forecasters in October 2017 declined to 1.8 percent from 2.3 percent from the previous survey.

Chart 2.2 Food-in-core inflation stabilized at low level in line with low LPG and fresh food prices



Note: Contribution shows decomposition of inflation changes according to weights of each goods in CPI basket
Source: Ministry of Commerce, calculation by Bank of Thailand

Chart 2.3 Nonfood in core inflation edged up following increase in excise tax on tobacco and alcoholic beverages



Note: Contribution shows decomposition of inflation changes according to weight of each good in CPI basket
Source: Ministry of Commerce, calculation by Bank of Thailand

Table 2.1 Inflation

Annual percentage change	2016		2017				2018
	Q3	Q4	Q1	Q2	Q3	Q4	Jan-Feb
Headline Consumer Price Index (Headline CPI)	0.26	0.69	1.25	0.10	0.45	0.88	0.56
Core Consumer Price Index (Core CPI)	0.76	0.73	0.66	0.47	0.49	0.61	0.61
Raw food	2.58	1.54	0.61	-2.99	-2.25	-0.80	-1.23
Energy	-7.00	-1.06	6.69	2.67	4.86	5.24	2.62

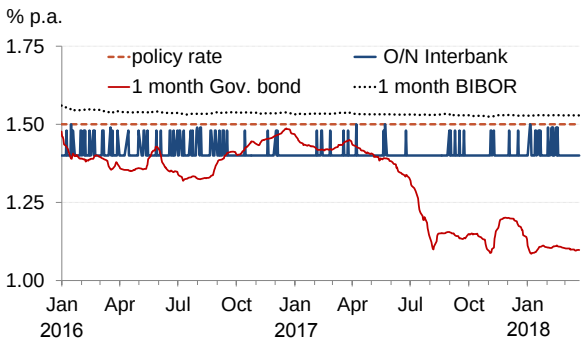
Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term money market rates remained low, while long-term government bond yields declined at the beginning of the year due to higher demand from investors and subsequently gradually increased primarily on account of external factors.

Most **short-term money market rates** remained close to the policy interest rate in the first quarter of 2018. However, short-term government bond yields declined at the beginning of the year and stabilized at the level below the policy rate. This was attributable to a higher demand from foreign investors in tandem with the appreciating trend of the baht and low borrowing costs of the baht in the foreign exchange swap market, coupled with a limited supply of short-term bonds issued by the Bank of Thailand (Chart 2.4). Meanwhile, medium- and long-term government bond yields fell at the beginning of the year on account of higher demand from both domestic and foreign investors, partly due to portfolio adjustments at the beginning of every year. After that, yields gradually rose in line with the increase in U.S. government bond yields following better-than-expected U.S. economic and inflation outturns. However, Thai bond yields increased to the lesser degree because (1) the net supply of bonds remained low given a large amount of matured bonds in the first quarter, resulting in high investor demand relative to supply, (2) Thai bonds were still attractive relative to those of other regional countries, leading to continued fund flows from foreign investors into the Thai bond market, and (3) with Thailand's inflation expectations remaining low, investors therefore expected Thai government bond yields to remain low. Nonetheless, concerns over U.S. foreign trade policies and potential retaliatory measures from other countries did not pose any significant impact on Thai government bond yields (Chart 2.5).

Chart 2.4 Short-term interest rates remained close to policy rate, except short-term government bond yields which declined at beginning of the year and stabilized at low levels

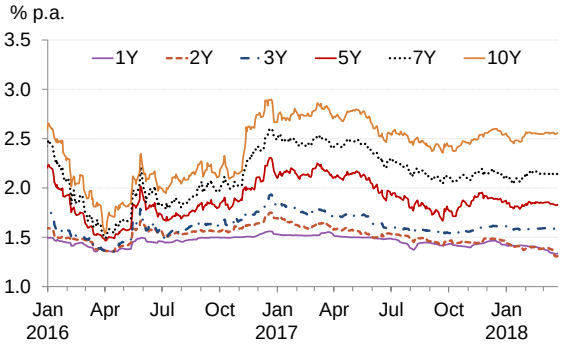
Short-term rates in financial markets



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 2.5 Movements in medium- and long-term government bond yields were driven by higher demand from foreign investors and external factors

Government bond yields



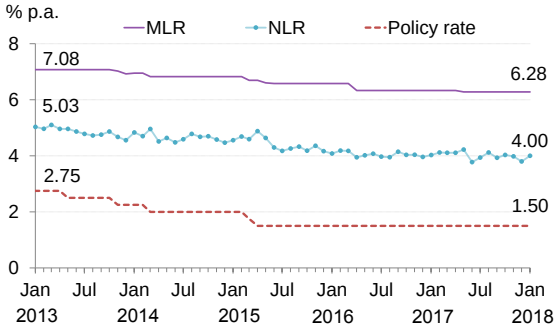
Source: Thai Bond Market Association (Thai BMA)

Corporate bond yields trended down in tandem with smaller credit spreads. Meanwhile, financing costs through commercial banks, as reflected in the new loan rate (NLR)[2/], remained at a low level, indicating financial conditions that were accommodative (Chart 2.6).

Private credit extended to both businesses and households continued to expand, together with improvements in credits extended to SMEs observed in several businesses, reflecting a more broad-based economic recovery.

Chart 2.6 New Loan Rate (NLR) stabilized at low level

New Loan Rate



Source: Bank of Thailand

Private credit [3/] accelerated 4.3 percent in February 2018 from the same period last year (Chart 2.7), higher than 4.1 percent observed in the fourth quarter of 2017. Business credit continued to expand on account of loans extended to large corporates in food and beverage sectors mainly to acquire foreign businesses. Meanwhile, loans to SMEs improved in several businesses such as wholesale food and beverages, real estate, rubber and plastic products, and warehouses. Such improvement reflected a more broad-based economic recovery. In addition, household credit accelerated across all loan purposes, particularly auto leasing which continued to accelerate in line with demand expansion and new car launches.

The net issuance of corporate bonds increased 9.0 percent in the fourth quarter of 2017 from the same period last year. This was particularly observed in finance and banking, wholesale and retail, food and beverages, and real estate sectors. Net issuance of corporate bonds was relatively stable in February 2018 (Chart 2.8), where issuances of such bonds were due to funding in many businesses such as in the financial, energy, transport, real estate, and

[2/] NLR is calculated based on a weighted average of interest rates for new loan contracts extended by 14 Thai commercial banks (excluding consumer loans and loans to financial intermediaries). The data covers loans of value of 20 million baht or higher for all purposes and terms, and includes both secured and non-secured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract

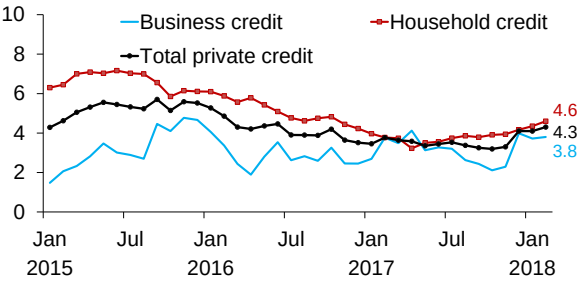
[3/] Outstanding credit of other depositary corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and market mutual funds.

food and beverage sectors. Funding through the equity market continued to increase in the fourth quarter of 2017, especially in businesses related to food, technology and agriculture. In February 2018, funding through the equity market increased slightly by businesses related to energy and real estate sectors. Overall, funding in the private sector continued to expand albeit at a slower pace.

Chart 2.7 Private credit, extended to both businesses and household, continued to expand

Growth of private credit

Percentage change from the same period last year



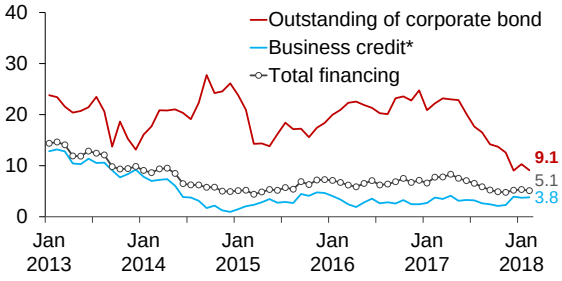
Note: Private credit includes credit to other depositary corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and money market mutual funds

Source: Bank of Thailand

Chart 2.8 Overall financing continued to expand albeit at slower pace

Growth of corporate bond outstanding and business credit

Percentage change from the same period last year



Note: *Business credit covers lending activities of Other Depository Corporations (ODCs)

**In August, businesses in telecommunication and energy sectors did not roll over the matured bonds.

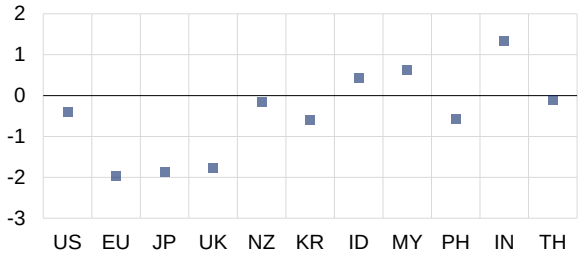
Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Going forward, financial conditions were expected to remain accommodative as reflected in the real policy interest rate which remained at a low level and was moderate compared with other countries (Chart 2.9). Meanwhile, financing costs through commercial banks, as reflected in the new loan rate (NLR), were expected to stabilize at a low level. Nevertheless, the Credit Condition Survey[4/] indicated that financial institutions would still maintain caution in extending credit in the first quarter of 2018, except for auto leasing where financial institutions became more vigilant.

Chart 2.9 Thailand's real policy rate* remained low and was moderate compared with other countries

Real policy rates*

Percent



Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to a survey by Consensus Economics (as of 12 March 2018)

Sources: Bloomberg, Consensus Economics, calculation by Bank of Thailand

The baht appreciated against the U.S. dollar, while the baht was somewhat stronger relative to movements of trading partners' currencies consistent with Thailand's economic fundamentals.

In the first quarter of 2018 **the baht appreciated against the U.S. dollar** relative to the end of previous quarter. The baht appreciation was mainly due to the weakening of the U.S. dollar in January (Chart 2.10). This was underpinned by the global economic outlook, particularly better economic outlooks of the euro area and Asia relative to the U.S., together with negative market sentiments on the U.S. dollar due to uncertainties surrounding U.S.

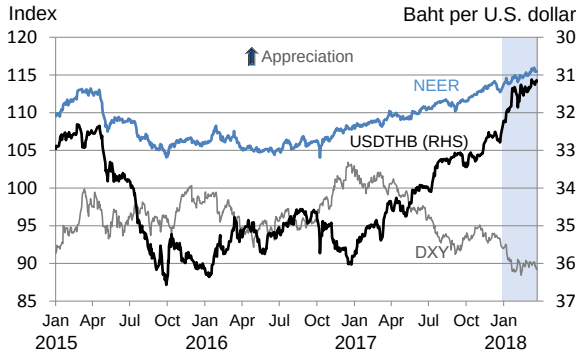
4/ Survey of credit conditions for the fourth quarter of 2017 and outlook for the first quarter of 2018.

foreign trade policies and concerns over the passage of the drafted budget bill which resulted in a partial government shutdown in the U.S. After that, the baht was relatively stable against the dollar, despite a sharp depreciation during a short period of global market corrections in February. Later in March, uncertainties pertaining to U.S. foreign trade policies heightened following the announcement of higher import tariffs on steel and aluminum by the U.S. This could affect confidence on the U.S. economy if there were retaliatory measures from major trading partners affected by the new tariffs such as China. Consequently, the baht would appreciate against the U.S. dollar in the same direction as most regional currencies, although the pace of appreciation varied depending on country-specific factors. As of March 27, 2018, the baht closed at 31.17 baht per U.S. dollar, appreciated 4.5 percent from the end of the previous quarter.

The nominal effective exchange rate (NEER) stood at 115.47 on March 27, 2018, a 2.3 percent appreciation from the end of the previous quarter, as the baht appreciated against most trading partner currencies (Chart 2.11). As of the end of February 2018, the real effective exchange rate (REER) rose 2.0 percent from the end of last year, which was close to NEER appreciation. On the whole, REER appreciation was consistent with Thailand's economic fundamentals especially its strong external stability. In the period ahead, exchange rates would likely remain volatile due to uncertainties surrounding monetary, fiscal, and foreign trade policies of major advanced economies.

Chart 2.10 The baht strengthened against U.S. dollar due to dollar weakening

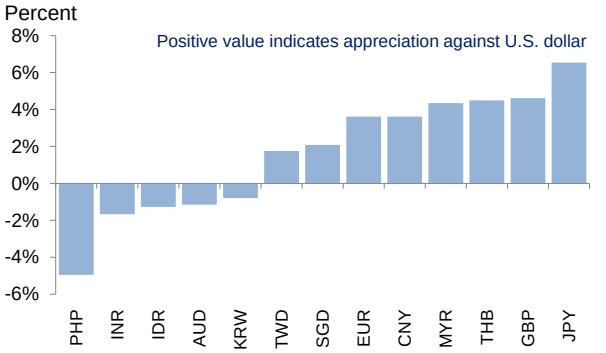
USDTHB, NEER, DXY



Sources: Bank of Thailand and Reuters (data as of 27 March 2018)

Chart 2.11 Major currencies and most regional currencies appreciated against U.S. dollar

Currency changes against U.S. dollar (27 Mar 18 compared to 29 Dec 17)



Sources: Bank of Thailand and Reuters (data as of 27 March 2018)

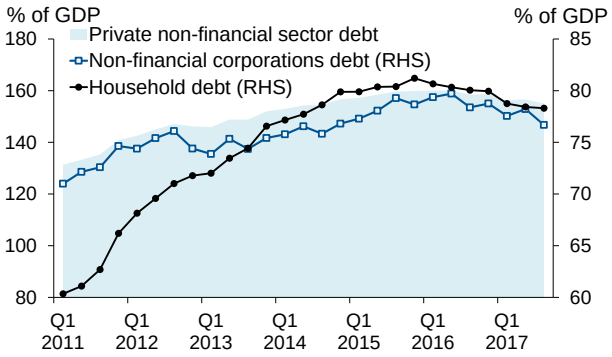
Financial stability remained sound overall, with continued economic expansion leading to improvements in financial positions of the private sector. However, there was increased accumulation of risks in certain pockets that warranted monitoring, particularly debt serviceability of households and some SMEs, the continued search-for-yield behavior, and an oversupply of property in the real estate market in certain price ranges and areas.

Thailand's financial stability remained sound overall with strong external stability as reflected in a high level of foreign exchange reserves and high liquidity of foreign currencies, given sustained current account surplus which would provide cushion against volatilities in global financial markets, and a low level of external debt to GDP relative to emerging economies. In addition, financial positions of financial institutions remained strong as reflected in high levels of capital buffers and provisions for loan losses to cushion against risks stemming from deteriorating credit quality. Risks related to leveraging of the private sector decreased overall, as reflected in the downward trend of debt to GDP of non-financial

corporates (Chart 2.12). This was on account of a continued economic expansion and a smaller growth of corporate and household debts relative to the previous 2-3 years. However, the Committee assessed that there remained pockets of risks to be monitored going forward summarized as follows.

Chart 2.12 Private debt-to-GDP ratio trended down since second half of 2016

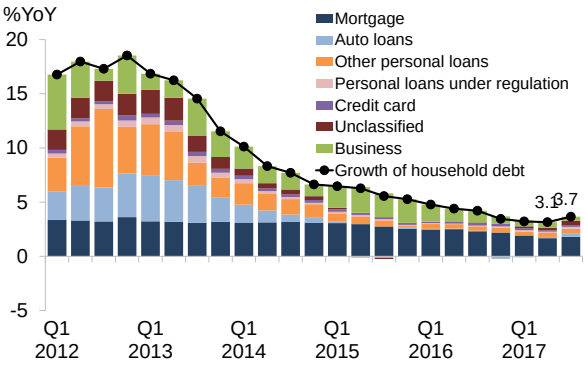
Private non-financial sector debt-to-GDP ratio



Source: Bank of Thailand

Chart 2.13 Household debt slightly increased after having slowed down in earlier period

Contribution to growth of household debt by loan purpose



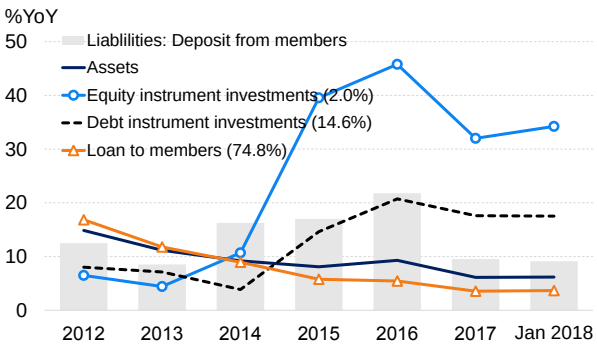
Source: Bank of Thailand

(1) Household debt showed signs of a slight acceleration while debt serviceability of certain households deteriorated. The latest data as of the third quarter of 2017 indicated that household debt accelerated slightly in almost all loan purposes, particularly credit card and auto loans which grew in line with economic expansion and new debt accumulation following an expiration of the first-car scheme (Chart 2.13). Meanwhile, debt serviceability continued to warrant monitoring, although the non-performing loan (NPL) ratio for consumer loans declined to 2.68 percent in the fourth quarter of 2017 from 2.74 percent in the previous quarter. Such decrease was partly attributable to end-of-year consumer loan portfolio management of financial institutions. However, a rise in new-entry NPLs was observed, especially in mortgage loans.

(2) Some SMEs which continued to incur losses would see deterioration in their abilities to service debt and cushion against economic volatilities. Performances and financial positions of some businesses, especially SMEs, remained fragile, as reflected in the operating profit margin (OPM) and interest coverage ratio (ICR) of small businesses in the fourth quarter of 2018[5] that continued to be negative. This was consistent with a high NPL ratio of 4.4 percent for SME loans.

Chart 2.14 Investment in equity instruments of savings cooperatives began to slow down

Asset, investment, and deposit growth of savings cooperatives



Note : ( ) denotes share to total assets of savings cooperatives as of December 2017

Source: Cooperative Auditing Department, calculation by Bank of Thailand

(3) The continued search-for-yield behavior could lead to underpricing of risks. A prolonged period of low interest rates was one of the key factors prompting investors to invest in riskier assets. Despite limited

[5/] calculation of ICR and OPM at 25th percentile of small-sized companies listed at the Stock Exchange of Thailand

systemic risk overall, there were pockets of risks that warranted close monitoring. Such risks include (1) overseas investment through foreign investment funds (FIF) that remained high and could lead to concentration risk since these investments were mainly concentrated in only five countries, and (2) a continued search-for-higher-yield behavior through savings cooperatives, whose assets and deposits increased at a high rate albeit slowing down somewhat after regulatory authorities collaborated to enhance supervisory standards (Chart 2.14). At the same time, accumulation of credit and liquidity risks also warranted monitoring as well as growing linkages, through increased borrowings, between savings cooperatives. In addition, most savings cooperatives continued to expand their investments in bonds and equities, thereby increasing the importance of savings cooperatives to the financial system. The Committee would therefore continue to closely monitor such linkages as well as developments of savings cooperatives.

(4) An oversupply in the real estate sector was observed in certain price ranges and areas. An oversupply of condominium units with price below 3 million baht continued to increase, while the time taken for all units to be sold appeared longer. Consequently, this could affect liquidity of some property developers, particularly small and medium-sized developers. In addition, an oversupply of condominium units in certain areas warranted monitoring, especially those along the MRT Purple Line (Khlong Bang Phai – Tao Poon). Although some developers responded by postponing launches of new projects and terminating some projects that were already open for sale, prevailing low demand meant that developers would take some time for all the excess supply of condominium units to be sold.

2.2 Outlook for the Thai Economy

Under the Committee's assessment, Thailand's economic growth was projected to gain further traction. The Thai economy would expand 4.1 percent in 2018, a higher growth rate than previously estimated in the previous *Monetary Policy Report*, and would continue expanding at the same rate in 2019. Key growth drivers included (1) a continued expansion of merchandise exports and tourism in line with trading partners' growth, (2) gradual improvements in private spending, and (3) benefits from government measures aimed at improving the well-being of low-income households which would partly help drive private spending. Meanwhile, inflation was projected to remain at a low level due mainly to supply-side factors but would slowly trend up.

Table 2.2 Forecast summary

Percent	2017*	2018	2019
GDP growth	3.9	4.1 (3.9)	4.1
Headline inflation	0.7	1.0 (1.1)	1.2
Core inflation	0.6	0.7 (0.8)	0.8

Note: * Outturn
() Monetary Policy Report December 2017
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies were projected to achieve higher growth than previously estimated. In particular, the U.S. economy would continue to expand on the back of domestic demand underpinned by the tax reform. This would consequently benefit exports of China, Asia, and the euro area whose exports were projected to achieve higher growth rate than previously expected.

- The federal funds rate was expected to be raised three times in each 2018 and 2019 and the Fed was expected to gradually commence its balance sheet reduction in accordance with the announced plan.

- Asian currencies (excluding the Chinese yuan) were stronger than the previous assessment throughout the forecast horizon given stronger-than-expected outturns in the first quarter of 2018 and improving economic fundamentals of Asian economies. In the period ahead, Asian currencies were expected to slightly appreciate, where the impact from increases in the federal funds rate on Asian currencies would likely be less than previously assessed.

- The Dubai crude oil price was revised up throughout the forecast horizon given higher-than-expected outturn. The increase was due to a decrease in crude oil supply toward a more balanced level on account of a rising demand for oil in line with global economic growth and the production cut in accordance with agreement between OPEC and non-OPEC producers.

- Farm income was revised down from the previous assessment due to a decline in agricultural prices as a result of a higher global agricultural output such as rubber, palm oil, and livestock, despite improvements in the prices of rice owing to a higher external demand.

- Public spending at current prices was revised down due to (1) a larger-than-expected impact from the Public Procurement and Supplies Management Act, B.E. 2560, particularly on delay in disbursement by local governments, which could result in a lower public expenditure on both consumption and investment, (2) the holdover of some investment projects of state-owned enterprises due to problems in accessing construction sites and delay in contract signing from the previous assessment, and (3) delay in the construction of three motorway projects, namely Bangyai–Kanchanaburi, Bang Pa In–Nakornratchasrima, and Pattaya–Mabtapud, due to difficulty in accessing construction sites and insufficient land expropriation budget.

Table: Summary of forecast assumptions

Annual percentage change	2017*	2018	2019
Dubai crude oil price (U.S. dollar per barrel)	53.1	62.4 (55.0)	63.0
Farm income (% YoY)	1.7	3.1 (4.1)	2.2
Government consumption at current price (billion baht)[1/]	2,532	2,676 (2,708)	2,838
Public investment at current price (billion baht)[1/]	926	1,038 (1,067)	1,092
Fed funds rate (% at year end)	1.38	2.13 (2.13)	2.88
Trading partners' GDP growth (% YoY)[2/]	3.9	3.7 (3.5)	3.6
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)[3/]	155.7	148.0 (154.3)	147.0

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report December 2017

Merchandise exports were expected to continue expanding across various product categories and almost all export destinations.

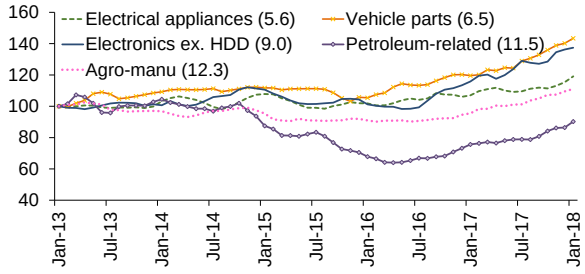
The value of merchandise exports was projected to continue expanding and record 7.0 and 3.6 percent growth in 2018 and 2019, respectively, where growth would be observed both in terms of price and volume. Export prices were expected to trend up in line with crude oil prices, especially for commodities and oil-related products such as petroleum products. Export volume was expected to gain further traction across various product categories, particularly electronics, automobile parts, and processed agricultural products, (Chart 2.15) and in almost all export destinations (Chart 2.16) thanks to expansion of the global economy and global trade volume. Moreover, Thai exports of electronics were projected to record robust growth, especially integrated circuits which would be used in Internet of Things (IoT) devices and electronic parts in automobiles. **However, the value of merchandise exports would likely register a slower growth in 2019,** consistent with a slowdown in trading partners' growth and global trade volume, after having expanded at a higher growth rate than historical average in the previous period. In addition, Thai exports would also be constrained by structural problems in certain industries which would take time to be resolved.

Nevertheless, the Committee viewed that the increase in import tariffs by the U.S. on certain products was expected to have a limited direct impact on Thai merchandise exports in the short term, as only a small share of Thai products in total Thai exports would be subject to these new tariffs. However, the Committee would continue to monitor the potential indirect effect of the increase in import tariffs by the U.S. on the global economy, particularly through supply chains, which could weigh on international trade and Thai merchandise exports in the period ahead.

Chart 2.15 Merchandise exports continued to expand across various product categories

Value of merchandise exports, by product category

Seasonally adjusted index, 3-month moving average (January 2013 = 100)



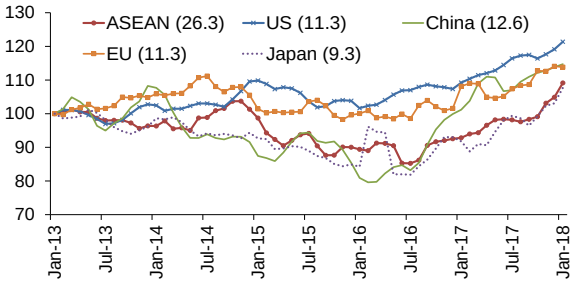
Note: Number in () denotes share in total exports in 2017

Source: Customs Department, calculation by Bank of Thailand.

Chart 2.16 Merchandise exports expanded across almost all export destinations

Value of merchandise exports by export destination

Seasonally adjusted index, 3-month moving average (January 2013 = 100)



Note: Number in ( ) represents share in total exports in 2017

Source: Customs Department, calculation by Bank of Thailand

Exports of services would continue to expand on the back of the tourism sector.

Thailand's exports of services were expected to expand in line with continued improvement in tourism. The projection of the number of foreign tourists was revised up to 37.6 million in 2018 from the previous forecast of 37.3 million, while the number of foreign tourists would likely rise to 39.0 million in 2019. Key factors supporting the tourism sector included a steady growth in the number of Chinese tourists, both group tourists with higher spending per head and free and independent travelers (FIT) with high purchasing power. In addition, Indian tourists would likely increase after the Airports Authority of India

granted airlines registered in Thailand additional capacity entitlements under bilateral air services agreement since March 2018. However, growth of Thailand's tourism sector would likely slow down given capacity constraints in accommodating the rising number of tourists in the main airports and the delay in the Airport of Thailand Plc (AOT)'s Suvarnabhumi Airport development project phase 2, which was under the standard price review. Nevertheless, such constraints were eased somewhat with increasing roles of local airports operated by the Department of Airports and the Ministry of Defense, especially Krabi, Suratthani, and U-tapao Airports, in facilitating international flights.

Given improvements in the value of merchandise and services exports, the projection for the value of merchandise and services imports was revised up with higher imports of raw materials and intermediate goods and higher oil prices. Consequently, **the current account would likely record a slightly smaller surplus than estimated in the previous *Monetary Policy Report*, registering 42.2 billion U.S. dollar in 2018 and was expected to fall to 39.5 billion U.S. dollars in 2019**.

Private consumption would gradually expand.

Private consumption would gradually expand supported by earnings of non-agricultural medium-to-high income households that remained sound. In addition, earnings of low-income households began to show signs of improvement as employment picked up in export-related businesses and the tourism sector. Meanwhile, household income in some agricultural sectors was projected to slow down in line with agricultural prices. Overall household purchasing power would gradually improve. This was partly a result of structural changes in the labor market such as adoption of automation in place of human labor in the production process as well as a decline in agricultural prices which made farming unattractive. Hence, these led to a greater migration of workers out of the manufacturing and agricultural sectors to the services sector. Workers who migrated from the manufacturing sector to the services sector might earn less as they mostly migrated to certain parts of the services sector with relatively lower productivity and wages. Moreover, elevated household debt underlined the fact that workers still needed to earmark part of their income for debt repayment. However, government measures, such as the social welfare card project (second phase), the community enterprise development project and the agricultural reform project, would partly help raise consumption of low-income households in the period ahead (Box: Assessing private consumption recovery).

Public spending helped drive growth to the lesser extent than previously assessed.

Public spending partly helped drive economic growth on the back of continued expansion in public consumption and investment. However, public expenditure was lower than assessed in the previous *Monetary Policy Report*. This was because the impact from the Public Procurement and Supplies Management Act, B.E. 2560 on government disbursement was larger and longer than expected. In particular, disbursement by certain state agencies was significantly affected, especially local governments that would now be governed by the new law and had not previously operated under this system. Those agencies would thus need some time to adjust. Moreover, there were delay in the construction of three motorway projects including Bang Yai–Kanchanaburi, Bang Pa In–Nakhon Ratchasima and Pattaya–Map Ta Phut, due to difficulty in accessing construction sites and insufficient land expropriation budget.

Meanwhile, most investment projects of state-owned enterprises (SOEs) were not affected by the new law[6/] and were mostly on track. However, some projects were postponed such as the second phase of the Mass Rapid Transit Authority (MRT)'s Orange Line project (Thailand Cultural Centre–Min Buri), the State Railway of Thailand's five double-track railway projects, and the National Housing Authority's housing development project.

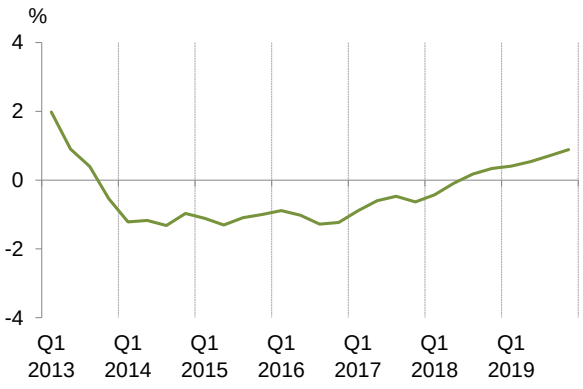
Private investment would continue to improve.

Private investment would continue to improve consistent with growth of exports and private consumption. Such improvement was partly reflected by a rising demand for business credit and the increase in the total value of investment applications submitted to the Board of Investment (BOI) at the end of 2017. Moreover, investment incentives offered by the government would play an important role in driving private investment, particularly through public-private partnership (PPP), public investment in infrastructure projects, and development projects under the Eastern Economic Corridor (EEC) initiative whose prospects became more certain. Therefore, these measures would help promote business confidence and foster an investment-friendly environment that would attract greater foreign investment. Apart from these, government measures aimed at improving the well-being of low-income households—such as the community enterprise development and agricultural reform projects—would partly help boost domestic investment to some extent.

Inflation stabilized at a low level in the short run but was expected to slowly rise.

In recent periods, inflation stabilized at a lower level than previously estimated in the previous *Monetary Policy Report*. This was owing to continued contraction in fresh food prices following declined prices of meat, particularly pork, stemming from a substantial increase in supply, and a fall in prices of vegetables and fruits given higher agricultural output thanks to favorable weather conditions. In addition, prices did not sufficiently accelerate as they were held back by a gradual expansion in domestic demand.

Chart 2.17 Output Gap



In the period ahead, inflation was projected to rise at a slower pace than previously assessed in the previous *Monetary Policy Report*. This was attributable to a slower increase in cost-push inflationary pressures relative to the recent periods, as costs of fresh food prices would likely remain low due to the government's water resource management aiming to improve drainage for agriculture purposes. Other contributing factors included structural changes from technological advancements that helped reduce costs of goods and services and intensifying business competition. Meanwhile, the impact from the minimum wage increases in 2018 on inflation was largely limited, as only a

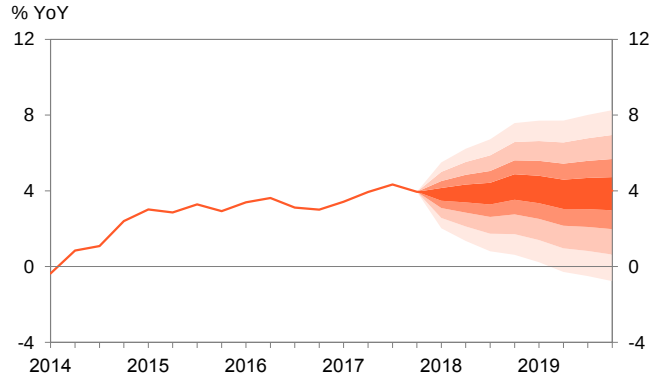
[6/] Procurement of state-owned enterprises (SOEs) directly related to commerce was exempt from such law. Also, on January 31, 2018, the Public Procurement and Supplies Management Policy Committee approved a draft of the new procurement practices for 17 SOEs. Hence, this would allow spending by SOEs to continue smoothly.

small share of workers in total labor force received higher minimum wages. Therefore, the impact from the minimum wage increases on the overall wage was assessed to be relatively small. Meanwhile, **demand-pull inflationary pressures** were close to the previous estimates. Although the output gap was expected to close in the latter half of 2018 (Chart 2.17), inflation might not increase significantly as demand-pull pressures would be constrained by positive spillovers from economic expansion that were not sufficiently broad-based and continuous sales promotion offered by businesses. **The Committee therefore projected headline inflation to average 1.0 and 1.2 percent in 2018 and 2019 respectively, while core inflation was projected to average 0.7 and 0.8 percent in 2018 and 2019 respectively**.

Risks to the growth projection were expected to tilt to the downside.

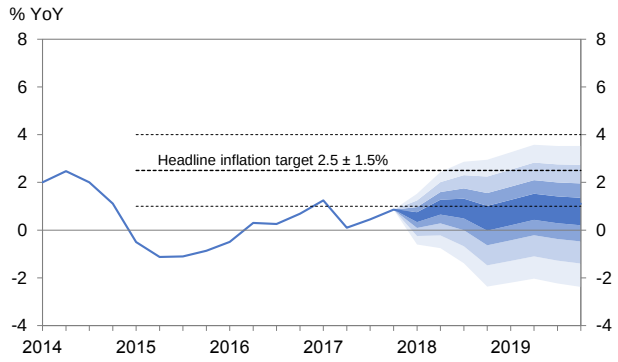
Under the Committee's assessment, **risks to the growth projection were expected to tilt to the downside**, as reflected in the fan chart that skewed downward throughout the forecast horizon (Chart 2.18). **The Thai economy would face greater downside risks** stemming from both external and domestic factors. Risks pertaining to U.S. foreign trade policies and potential retaliatory measures by major economies could weigh on trading partner growth to a larger extent than previously assessed despite some improvements in geopolitical risks related to the Korean Peninsula. Moreover, domestic risks included a larger-than-expected impact from the Public Procurement and Supplies Management Act, B.E. 2560 on public expenditure and a lower-than-expected domestic spending as improvement in purchasing power was not yet sufficiently broad-based. However, on the upside**, there were possibilities that the growth outturn might be higher than the baseline projection** given a better-than-expected growth outlook of Thailand's trading partners with supporting factors including the U.S. tax reform, the Chinese government's measures which could soften its economic slowdown, as well as government infrastructure investment which could stimulate a higher private investment than the baseline projection. With regard to inflation, the Committee assessed **risks to both headline and core inflation projections would tilt to the downside** (Chart 2.19 and 2.20) consistent with risks to the growth forecast.

Chart 2.18 Growth forecast



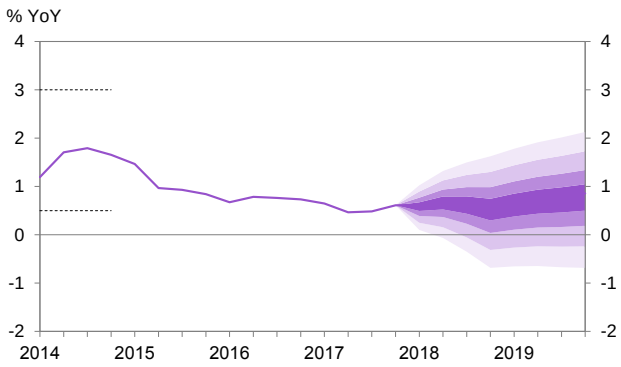
Note: Fan chart covers 90% of probability distribution

Chart 2.19 Headline inflation forecast



Note: Fan chart covers 90% of probability distribution

Chart 2.20 Core inflation forecast



Note: Fan chart covers 90% of probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2017*	2018	2019
GDP growth	3.9	4.1 (3.9)	4.1
Domestic demand	2.1	3.6 (3.4)	3.4
Private consumption	3.2	3.3 (3.1)	3.3
Private investment	1.7	3.0 (2.3)	3.6
Government consumption	0.5	2.9 (3.2)	3.3
Public investment	-1.2	9.5 (9.0)	3.4
Exports of goods and services	5.5	5.2 (3.7)	3.6
imports of goods and services	6.8	5.5 (3.5)	3.7
Current account (billion, U.S. dollars)	49.3	42.2 (43.1)	39.5
Value of merchandise exports	9.7	7.0 (4.0)	3.6
Value of merchandise imports	14.4	11.5 (7.5)	5.4
Number of foreign tourists (million person)	35.4	37.6 (37.3)	39.0

Note: *Outturns

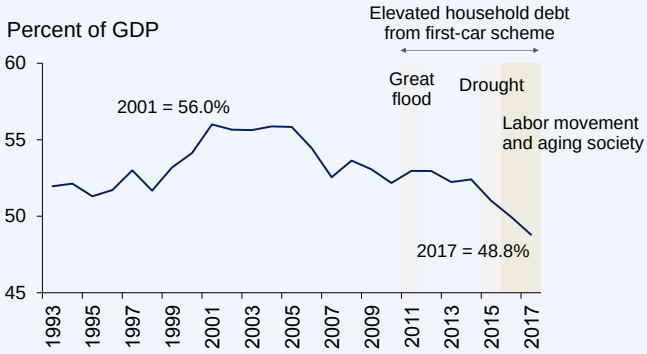
( ) Monetary Policy Report December 2017

Assessing private consumption recovery

Private consumption is considered one of Thailand's key growth drivers with its share of around half the country's gross domestic product (GDP). However, the role of private consumption has been declining, as reflected in the share of private consumption to GDP falling from its peak at 56.0 percent in 2001 to 48.8 percent in 2017 (Chart 1). Moreover, private consumption growth tends to stay below and diverge further from GDP growth particularly in the last five years despite their similar growth rates in the past (Chart 2). **The growth divergence could reflect both structural and cyclical changes in private consumption** where the divergence has been clearly observed since 2011. Since then, Thailand faced **series of economic structural changes**. First, **the great flood in 2011** undermined employment and household income in the affected areas as well as overall consumer confidence. In addition, some households bore an extra burden from restoration and reconstruction expenses on damaged houses and assets. Second, **household debt has remained persistently high since the 2011** and after the first-car[7/] scheme. Third, **severe drought in 2015** weighed on farm income. Fourth, there were **structural changes in the labor market** as workers moved from the manufacturing sector with higher average income to the services sector with lower average income. As a result, overall non-farm income continued to fall. Fifth, **Thailand's aging society** has prompted some consumers to save more for their retirement. As a consequence, household disposable income has consistently declined. (Chart 3) All of these factors have caused private consumption growth to be below its historical growth and GDP growth.

Chart 1 Private consumption to GDP ratio has fallen since 2011

Private consumption to GDP



Source: NESDB, calculation by Bank of Thailand

Chart 2 Private consumption growth is now below GDP growth

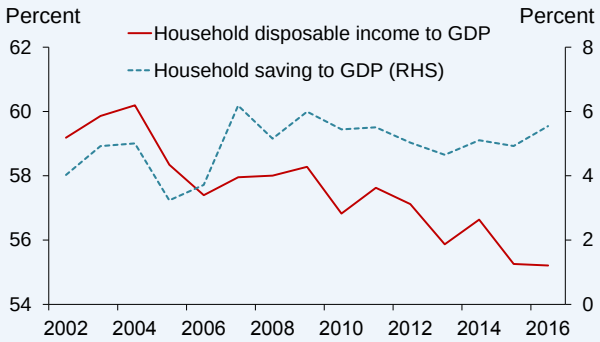
Real GDP and real private consumption growth

Period	Growth from the same period last year (percent)	
	Real GDP	Real Private Consumption
1990-2000	5.1	5.1
2001-2010	4.6	4.2
2011-2017	3.1	2.7
2013-2017	2.8	2.0

Source: NESDB, calculation by Bank of Thailand

Chart 3 Aging prompts consumers to save more for retirement; household disposable income declines

Household disposable income to GDP and household saving to GDP



Note: Household disposable income = corporate profits + labor compensation + net income from asset (rent & interest rate) + benefits from social security + transfer receipts - taxes on income and asset - social security payments - transfer payments

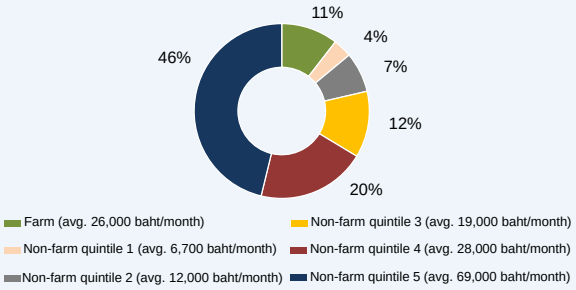
Sources: Bank of Thailand, NESDB, calculation by Bank of Thailand

[7/] Repayment of auto leasing and hire purchase loans typically takes place over 4-7 years.

Private consumption growth in recent periods is mainly attributable to spending by a high-income group. Middle and high-income non-farm households[8/] account for around 66 percent of overall spending (Chart 4). At the same time, income indicators for high-income non-farm households have steadily trended up[9/] (Chart 5). These leads to a continued expansion in consumption on services and durable goods, especially vehicles. On the other hand, consumption on semi-durable and non-durable goods, particularly consumer products, grows only slowly. This is partly because purchasing power of a low-income group—both in farm and non-farm sectors—has yet to fully recover. As a result, overall private consumption growth[10/] has remained low.

Chart 4 Private consumption expands mainly on the back of spending by middle- and high-income groups (around 66 percent of total expenditure)

Household expenditure structure by income group (average over Q1-Q3 2017)

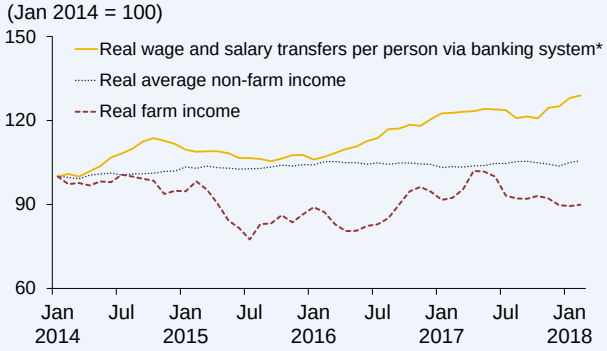


Note: Household income groups are classified by quintile (20%) of socio-economic survey (SES) in 2017, consisting of farm households 15% and non-farm households 85% of total households respectively.
Source: Socio-economic survey (SES)

Chart 6 Non-farm households in high-income group tend to have higher earnings relative to other groups, as reflected in real wage and salary transfers per person via banking system

Household income indicators

Index, seasonally adjusted (3-month moving average)
(Jan 2014 = 100)



Note: Wage and salary transfers are calculated from (1) transfer transactions reported by commercial banks to Bank of Thailand which covers 90% of all retail transfer transactions and (2) transfer transactions through Interbank Transaction Management and Exchange (ITMX) which covers 10% of all retail transfer transactions.
Sources: Bank of Thailand, Office of Agricultural Economics, National Statistical Office, Ministry of Commerce, calculation by Bank of Thailand

Private consumption cycles

Private consumption continues to gain further traction as reflected in an expansion phase of the cycle primarily led by durable goods consumption in line with Thailand's business cycle (Chart 6).[11/] Decomposition of the overall private consumption cycle suggests the following key findings (Chart 7). First, the durable goods consumption cycle is in an expansion phase, consistent with business cycle, reflecting that households will consume more on durable goods relative to normal trend given improved consumer confidence on the overall economy and future income. Second, the services consumption cycle is in an expansion phase. Third, the semi-durable goods consumption cycle is in a slow expansion phase. Fourth, the non-durable goods consumption cycle is in line with normal trend. Details of the analysis are as follows.

8/ Non-farm households whose income are in fourth and fifth quintiles

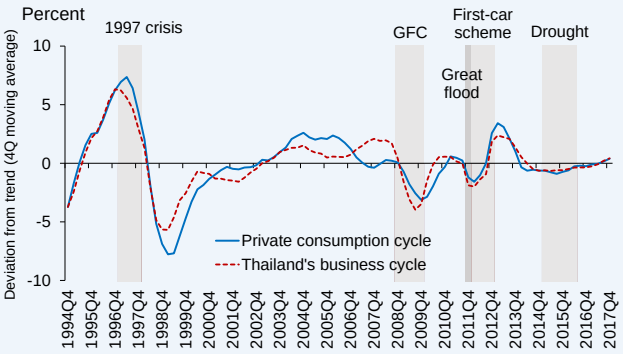
9/ Transfer payments of income and wages in baht term per person per month are indicators of income for high-income households

10/ Largest share of private consumption contributes to consumption on services (e.g. rent and transportation), non-durable goods (e.g. food), durable goods (e.g. vehicles) and semi-durable goods (e.g. apparels), respectively.

11/ Cycle analysis is a common assessment which detrends, seasonally adjusts, and corrects for irregularity of economic variables. The decomposition can identify cyclical factors which can measures deviation from trend of economic variables.

Chart 6 Private consumption cycle is currently in expansion phase in tandem with overall business cycle

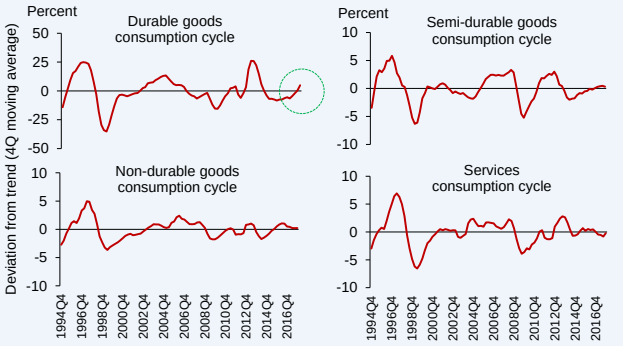
Private consumption cycle and overall business cycle



Source: NESDB, calculation by Bank of Thailand

Chart 7 Durable goods consumption cycle is in expansion phase due to declining debt burden from first-car scheme and increase in automobile demand due to launch of new models

Private consumption cycles classified by product categories



Source: NESDB, calculation by Bank of Thailand

The expansion phase of the **durable goods consumption cycle** is due to sustained income and confidence among the high-income group. In addition, declining debt burden from the first-car scheme and the launch of new car models will support consumption on durable goods especially vehicles going forward. The **services consumption cycle** is also in an expansion phase due to an upward trend in spending on tourism activities by Thais, both domestically and abroad. This will support consumption on services, particularly spending on hotels, restaurants, and transportation. The **semi-durable goods consumption cycle** is in a slow expansion phase partly supported by higher demand for spare and automobile parts following purchases of vehicles. Moreover, demand for textiles and apparels has shown an improving sign, as reflected in larger imported textiles and apparels in recent periods. Meanwhile, **consumption on non-durable goods** has only seen slight growth partly because purchasing power of low-income households—both in farm and non-farm sectors—was not yet sufficiently robust. However, the government has announced measures aimed at improving welfare of low-income households. These include the additional budget in 2018 which would be gradually spread in 2018 and 2019 through the social welfare card project (second phase), the community enterprise development project, as well as reforms in the agricultural sector with job creations during the drought season. Such measures are expected to support consumption of low-income households and more robust growth of overall private consumption going forward. Nonetheless, private consumption growth could somewhat fall below its historical growth rates due to structural changes in the labor market as more workers have moved to services sector and earned lower average income as well as Thailand's moving toward an aging society.

3. Monetary Policy Decision

In the context where the economy was projected to steadily gain traction, with inflation remaining low due partly to supply-side and structural factors, and with pockets of financial stability risks in a prolonged low interest rate environment, the Committee assessed the need for policy accommodation in the first quarter of 2018 with details summarized as follows.

Overall, the Thai economy was expected to steadily gain traction. In addition to exports, domestic demand would be additional growth driver, although it was not yet altogether sufficiently strong. Inflation was projected to edge up but at a slower pace than previously assessed due to supply-side factors. Risks to financial stability accumulated in certain areas. The Committee therefore had to weigh between sustaining economic growth in order to attain price stability and preserving financial stability. Key conclusions were as follows.

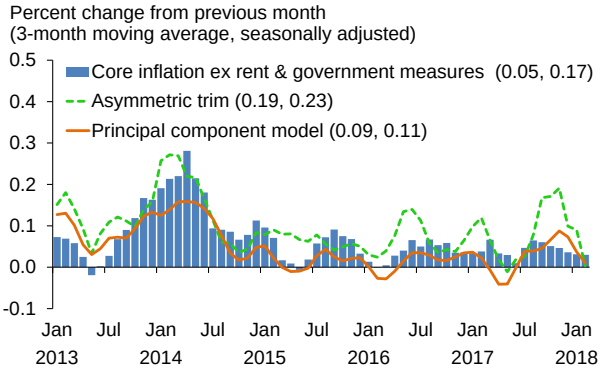
1. Pursuing sustainable economic growth. The Thai economy was expected to gain further traction where merchandise exports and tourism would continue to expand in line with stronger global economic growth; nonetheless, exchange rates could experience higher volatilities owing to external factors. Domestic demand gained additional momentum, particularly private spending which was supported by government measures, but it would expand only gradually. This was because overall economic growth had yet to fully benefit household income and employment, with the positive spillovers rather concentrated in the manufacturing sector and export-related businesses. Private consumption continued to expand due primarily to spending by middle and high-income household groups, while spending of low-income households did not fully recover. This was partly owing to declined agricultural prices, elevated debt levels, increased adoption of automation in place of human labor in the production process, and increased labor migration to the services sector with relatively lower wages. All these factors led to greater income disparities despite the overall improvement in the economy. Public expenditure remained a growth driver but there were risks of delayed disbursement which was partly due to the implementation of the Public Procurement and Supplies Management Act, B.E. 2560. However, the Committee viewed that monetary policy accommodation would help support the continuation of economic growth. The limited positive spillovers from economic expansion could be attributable to structural factors, in which case agricultural and labor reforms as well as targeted measures would be more effective than monetary policy. In addition, the Thai economy would face greater downside risks from the external front that warranted close monitoring. Such risks included in particular U.S. foreign trade policy and possible retaliatory measures by major economies.

2. Returning inflation to target in the medium term. A decline in inflation in recent periods reflected low inflationary pressures (Chart 3.1) due particularly to a decline in cost-push pressures because fresh food prices were expected to rise slower than previously assessed owing to the large supply in the market. However, headline inflation was projected to trend up in line with domestic demand growth and expected to return to target within the first half of 2018. In addition, inflation expectations based on surveys of businesses and professional forecasters became more stabilized (Chart 3.2). The Committee viewed that demand-pull inflationary pressures that remained low partly reflected domestic demand that was not yet strong. As a result, businesses faced constraints in raising prices. In addition, low

inflation pressures were attributable to structural changes such as technological developments and expansion of e-commerce which led to lower costs of production and intensified price competition. Thus, the Committee saw the need to closely monitor and assess changes in inflation dynamics, as this could significantly affect the pace of inflation target fulfillment and monetary policy conduct going forward.

Chart 3.1 Underlying inflation indicators edged down, reflecting low inflation pressures

Underlying inflation indicators

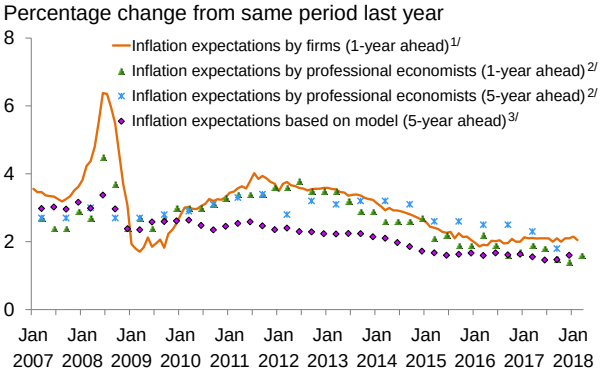


Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of February 2018, while the second value is 2004 - 2014 average; **Asymmetric trim** excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculation by Bank of Thailand

Chart 3.2 Short- and long-term inflation expectations became stabilized

Inflation expectations



Sources: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculation based on macro-finance term structure model with bond yield and macroeconomic data

3. Monitoring the buildup of financial stability risks. The Committee viewed that financial stability remained sound and with positive developments as reflected in private debt to GDP trending downward following economic growth. However, the buildup of vulnerabilities in certain areas continued and could result in abrupt adjustment in the financial system. Such vulnerabilities included, in particular, **debt serviceability of households and SMEs that continued to deteriorate as benefits from the economic expansion had yet to fully extend to these economic sectors.** NPLs of these groups remained high and household borrowing began to accelerate across almost all types of credit. Another vulnerability was **continued search for yield in a prolonged low interest rate environment that could lead to underpricing of risks.** This was reflected in a continued expansion of foreign investment funds (FIFs) and high deposit growth in saving cooperatives, especially those with larger asset sizes and greater linkages with the Thai financial system. The third vulnerability was an oversupply of property, especially condominiums along the skytrain and mixed-use real estate projects. Overall, the Committee assessed that these risks were manageable as over the recent periods regulatory authorities continued to enhance supervisory measures that prevent potential systematic risks. However, the Committee would closely monitor financial stability risks.

The Committee voted unanimously to keep the policy interest rate at 1.5 percent to maintain accommodative financial conditions in order to support a stronger economic growth and foster the return of inflation to target without causing the buildup of vulnerabilities to financial stability.

The Committee weighed various factors in formulating the most appropriate course of monetary policy and voted unanimously at the meeting on February 14, 2018 and later 6 to 1 at the meeting on March 28, 2018 to keep the policy rate unchanged at 1.50 percent. The Committee assessed that accommodative monetary policy would still be necessary to support a more robust growth in domestic demand which would foster the return of inflation to target and viewed that the policy rate at 1.50 percent would sufficiently facilitate accommodative financial conditions as reflected in low real interest rates. Moreover, New Loan Rate (NLR) that continued to trend down and bond yields that remained low led to sustained expansion of businesses financing through commercial banks and financial markets. Meanwhile, one Committee member voted to raise the policy rate by 0.25 percentage point to 1.75 percent at the meeting on March 28, 2018 as the unusually low interest rates for a prolonged period might prompt households and businesses to underestimate risks to potential changes in financial conditions. Furthermore, a gradual, timely reduction in monetary policy accommodation would not hinder economic growth or price stability; it would instead help alleviate risks to financial stability and facilitate a smooth process of monetary policy normalization.

With regard to exchange rates, the baht appreciated due to the weakening of the US dollar as the dollar depreciating against major currencies and most emerging economies currencies. However, the baht's appreciation was stronger relative to many regional currencies given Thailand's improved growth outlook as well as strong external positions as reflected in sustained current account surplus. The Committee viewed that such current account surplus was partly attributable to structural changes. For instance, a significant decline in oil prices since 2014 and China's policy that allowed greater outflows of Chinese tourists abroad had led trade and services balances to continue recording large surpluses. These structural changes had called for policy coordination among related parties in order for the balance of payments to be more sustainable. These policies included, for instance, the acceleration of public investment projects, especially projects with plans to import machinery and raw materials, and facilitation for both Thai individual and institutional investors to invest more abroad. These would partly help reduce pressures on the baht. Going forward, exchange rates would likely remain volatile due to uncertainties regarding the conduct of monetary and fiscal policies of major advanced economies. The Committee would thus continue to closely monitor developments in the foreign exchange markets and economic impacts.

Given the environment where central banks of major economies and many regional countries started to consider reducing the degree of monetary policy accommodation, the Committee viewed that appropriate monetary policy should be formulated based on a holistic view of the likely impacts on the Thai economy. Raising the policy interest rate when domestic demand still had to rely on support from government measures could weigh on demand-pull inflationary pressures and lead to risks to price stability and the anchoring of inflation expectations. In any case, with Thailand's external positions remaining sound, there was a limited risk of abrupt capital flow reversals that could be induced by differences in Thai and foreign interest rates. Meanwhile, risks to financial stability in connection with a prolonged low

interest rate environment were still manageable. Hence, monetary policy accommodation at the current policy interest rate could be maintained for some time.

Looking ahead, the Committee viewed that the current degree of monetary policy accommodation should be maintained for some time in order to support a stronger economic growth which could foster a gradual return of headline inflation to target in the medium term. The Committee would stand ready to utilize available policy tools to foster the return of inflation to target in a timely manner while seeking to achieve sustainable economic growth and financial stability.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2016	2017	2016	2017			
			Q4	Q1	Q2	Q3	Q4
GDP growth	3.3	3.9	3.0	3.4	3.9	4.3	4.0
Production							
Agriculture	-2.5	6.2	-0.4	6.0	15.9	9.7	-1.3
Non-agriculture	3.8	3.7	3.5	3.2	3.0	4.0	4.6
Manufacturing	2.3	2.5	2.9	1.9	1.0	4.2	3.0
Construction	8.6	-2.3	6.9	3.2	-5.7	-1.6	-5.3
Wholesales and retail trade	5.3	6.3	5.8	5.9	6.0	6.4	6.9
Hotels and restaurants	9.9	8.5	4.9	5.1	7.0	6.9	15.3
Transport, storage, and communication	4.1	7.3	3.7	5.3	7.8	7.4	8.9
Financial intermediation	6.5	5.0	7.6	4.8	6.3	4.6	4.2
Real estate, renting, and business activities	3.2	4.6	3.2	3.8	4.2	4.7	5.6
Expenditure							
Domestic demand	2.8	2.1	2.4	2.0	1.8	2.5	2.0
Private consumption	3.0	3.2	2.3	3.1	2.9	3.4	3.5
Private investment	0.5	1.7	-0.1	-1.1	3.0	2.5	2.4
Government consumption	2.2	0.5	2.9	-0.7	0.4	1.8	0.2
Public investment	9.5	-1.2	8.9	10.0	-6.9	-1.6	-6.0
Imports of goods and services	-1.0	6.8	3.5	5.9	7.2	6.5	7.5
imports of goods	-2.3	8.5	3.2	7.3	9.2	9.2	8.3
imports of services	4.6	-0.4	4.8	0.4	-1.0	-5.0	3.8
Exports of goods and services	2.8	5.5	1.8	2.7	5.1	6.9	7.4
exports of goods	0.3	5.6	1.6	2.8	4.9	8.2	6.6
exports of services	11.5	5.0	2.5	2.5	5.7	2.6	9.7
Trade balance (billion, U.S. dollars)	36.5	31.9	7.1	8.8	6.4	10.1	6.5
Current account (billion, U.S. dollars)	48.2	49.3	10.8	15.0	7.8	13.8	12.7
Financial account (billion, U.S. dollars)	-21.0	-19.2	-12.1	-7.0	-5.7	0.4	-6.8
International reserves (billion, U.S. dollars)	171.9	202.6	171.9	180.9	185.6	199.3	202.6
Unemployment rate (%)	1.0	1.2	1.0	1.2	1.2	1.2	1.1
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.1	1.1	1.1	1.2	1.3

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2016	2017	2017				2018	
			Q1	Q2	Q3	Q4	Jan	Feb
1. Financial market sector								
Bond market								
Bond spread (10 years - 2 years)	0.6	1.1	1.1	1.1	1.0	1.0	1.1	1.2
Equity market								
SET index (end of period)	1,542.9	1,753.7	1,575.1	1,574.7	1,673.2	1,753.7	1,721.4	1,697.4
Actual volatility of SET index[1/]	14.2	6.5	7.0	4.8	5.8	7.9	8.2	10.1
Price to Earnings ratio (P/E ratio) (times)	18.6	19.1	17.4	16.3	17.9	19.1	18.6	18.3
Exchange rate market								
Actual volatility of Thai baht (%annualized)[2/]	4.4	3.3	3.5	3.9	2.9	2.8	4.2	5.3
Nominal Effective Exchange Rate (NEER)	106.2	110.6	108.7	109.8	111.2	112.9	114.1	114.8
Real Effective Exchange Rate (REER)	100.6	103.6	102.7	102.8	104.2	105.7	106.0	n.a.
2. Financial institution sector[3/]								
Minimum Lending Rate (MLR)[4/]	6.26	6.20	6.26	6.20	6.20	6.20	6.20	6.20
12-month fixed deposit rate[4/]	1.38	1.38	1.38	1.38	1.38	1.38	1.38	1.38
Capital adequacy								
Capital funds / Risk-weighted asset (%)	18.0	18.2	17.8	17.9	18.4	18.2	n.a.	n.a.
Earning and profitability								
Net profit (billion, Thai baht)	199.0	187.3	51.2	49.0	46.5	40.8	n.a.	n.a.
Return on assets (ROA) (times)	1.1	1.05	1.2	1.1	1.0	1.1	n.a.	n.a.
Liquidity								
Loan to Deposit and B/E (%)	96.3	96.08	95.7	96.5	96.4	96.1	n.a.	n.a.
3. Household sector								
Household debt to GDP (%)	79.0	77.53922	77.9	77.5	77.3	77.5	n.a.	n.a.
Financial assets to debt (times)	2.6	2.7	2.7	2.7	n.a.	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Consumer loans	2.7	2.7	2.8	2.7	2.7	2.7	n.a.	n.a.
Housing loans	2.9	3.2	3.2	3.1	3.3	3.2	n.a.	n.a.
Auto leasing	1.8	1.6	1.6	1.7	1.6	1.6	n.a.	n.a.
Credit cards	3.7	2.6	4.1	3.4	2.8	2.6	n.a.	n.a.
Other personal loans	2.9	2.5	2.8	2.6	2.7	2.5	n.a.	n.a.
4. Non-financial corporate sector[5/]								
Operating profit margin (OPM) (%)	8.2	8.3	8.5	7.3	8.5	8.1	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.5	6.6	6.2	6.0	6.4	7.8	n.a.	n.a.
Current ratio (times)	1.6	1.7	1.7	1.6	1.6	1.7	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Large businesses	1.5	1.8	1.6	1.8	1.7	1.8	n.a.	n.a.
SMEs	4.3	4.4	4.5	4.4	4.6	4.4	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2016	2017	2017				2018	
			Q1	Q2	Q3	Q4	Jan	Feb
5. Real estate sector								
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)								
Total	61,452	62,664	12244	15086	16,859	18,475	3,639	4,408
Single-detached and semi-detached houses	13,409	13,907	2802	3544	3,774	3,787	708	1,061
Townhouses and commercial buildings	20,187	20,536	4315	4947	5,604	5,670	1,254	1,581
Condominiums	27,856	28,221	5127	6595	7,481	9,018	1,677	1,766
Number of new housing units launched for sale (Bangkok and Vicinity) (units)								
Total	110,575	114,477	28,924	25,256	35,434	24,863	4,112	4,161
Single-detached and semi-detached houses	19,433	14,280	2,968	2,830	4,874	3,608	899	1,580
Townhouses and commercial buildings	32,792	36,571	11,183	7,665	9,831	7,892	1,023	1,903
Condominiums	58,350	63,626	14,773	14,761	20,729	13,363	2,190	678
Housing price index (2009 = 100)								
Single-detached houses (including land)	130.8	130.9	128.6	129.6	131.6	133.9	136.4	138.1
Townhouses (including land)	137.6	141.2	138.3	140.0	142.6	143.7	144.4	145.9
Condominiums	166.2	171.0	169.8	168.8	169.8	175.4	178.6	181.6
Land	171.2	171.7	171.3	164.2	172.9	178.3	176.9	177.2
6. Fiscal sector								
Public debt to GDP (%)	40.8	41.2	41.7	41.3	41.9	41.2	41.4	n.a.
7. External sector								
Current account balance to GDP (%)[6/]	11.7	10.8	13.8	7.1	11.9	10.4	n.a.	n.a.
External debt to GDP (%)[7/]	32.5	35.2	33.2	34.0	35.5	35.2	n.a.	n.a.
External debt (billion, U.S. dollars)	132.2	149.0	136.2	140.3	148.1	149.0	155.9	153.7
Short-term (%)	41.2	41.9	40.5	39.4	40.9	41.9	41.7	41.5
Long-term (%)	58.8	58.1	59.5	60.6	59.1	58.1	58.3	58.5
International reserves / Short-term external debt (times)	3.2	3.2	3.3	3.4	3.3	3.2	3.3	3.3

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2018				2019			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 9	0	0	0	1	1	1	2	3
8-9	0	0	1	2	3	2	3	3
7-8	0	1	3	6	6	5	6	6
6-7	2	5	8	11	10	9	9	9
5-6	11	15	15	16	15	13	13	12
4-5	31	25	21	19	17	16	15	14
3-4	34	25	21	17	17	16	15	14
2-3	18	17	16	13	13	14	13	13
1-2	4	8	10	8	9	10	10	10
0-1	0	2	4	4	5	7	7	7
(-1)-0	0	1	1	2	3	4	4	4
(-2)-(-1)	0	0	0	1	1	2	2	2
(-3)-(-2)	0	0	0	0	0	1	1	1
< (-3)	0	0	0	0	0	0	1	1

Table: Probability distribution of headline inflation forecast

Percent	2018				2019			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 4.0	0	0	0	1	2	3	3	3
3.5-4.0	0	0	1	1	2	3	3	3
3.0-3.5	0	1	3	3	4	5	4	4
2.5-3.0	0	3	5	5	6	7	7	6
2.0-2.5	1	9	9	7	8	9	9	8
1.5-2.0	5	16	13	10	11	11	11	10
1.0-1.5	18	20	15	12	12	12	11	11
0.5-1.0	30	19	15	12	12	11	11	11
0.0-0.5	26	15	13	12	11	10	10	10
(-0.5)-0.0	14	9	10	10	9	9	9	9
(-1.0)-(0.5)	5	5	7	9	8	7	7	7
(-1.5)-(1.0)	1	2	4	7	6	5	5	6
(-2.0)-(-1.5)	0	1	2	5	4	3	4	4
< -2.0	0	0	2	7	6	5	6	7

Table: Probability distribution of core inflation forecast

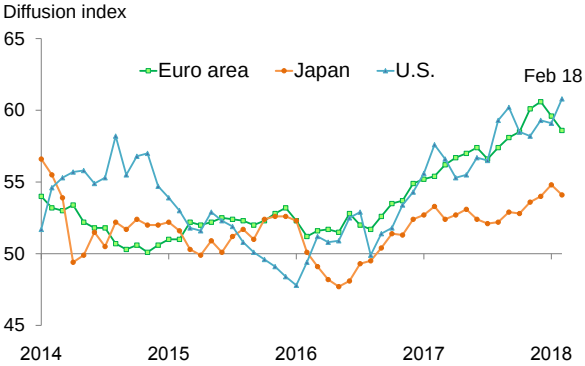
Percent	2018				2019			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	0	0
2.5-3.0	0	0	0	0	0	1	1	1
2.0-2.5	0	0	0	1	2	3	4	5
1.5-2.0	0	2	4	6	8	10	11	12
1.0-1.5	6	18	19	17	19	20	20	20
0.5-1.0	56	44	34	27	26	25	24	23
0.0-0.5	36	29	27	25	23	21	20	19
(-0.5)-0.0	2	6	12	15	14	12	12	11
(-1.0)-(0.5)	0	0	3	6	5	5	5	5
(-1.5)-(1.0)	0	0	0	2	2	2	2	2
(-2.0)-(-1.5)	0	0	0	0	0	0	0	0
< -2.0	0	0	0	0	0	0	0	0

4.2 Data pack

Global Economy

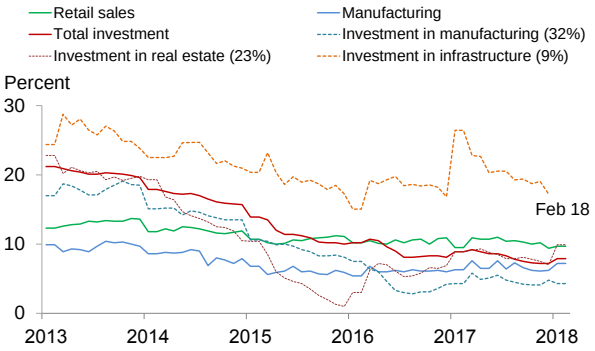
Thailand's trading partner economies continued to expand and remained a key driver of Thai exports going forward. Nonetheless, downside risks to the growth forecast increased from U.S. foreign trade policy. Meanwhile, inflation remained low and was expected to slowly rise. As a result, most central banks maintained monetary policy accommodative stance. However, some central banks gradually reduced monetary policy accommodation in line with improvements in growth and inflation outlooks.

Manufacturing Purchasing Manager Index



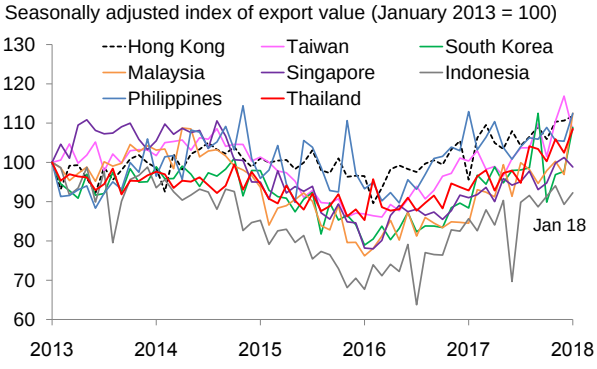
Sources: Bloomberg and Eurostat

China's economic indicators (Change from same period last year)



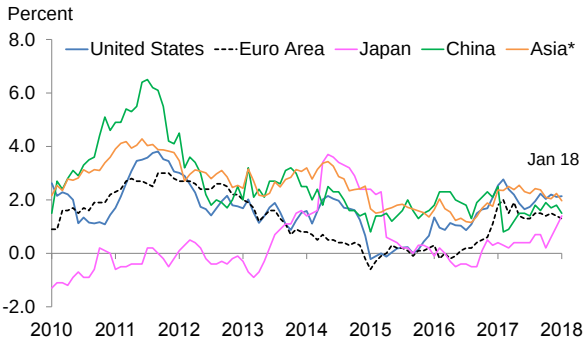
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

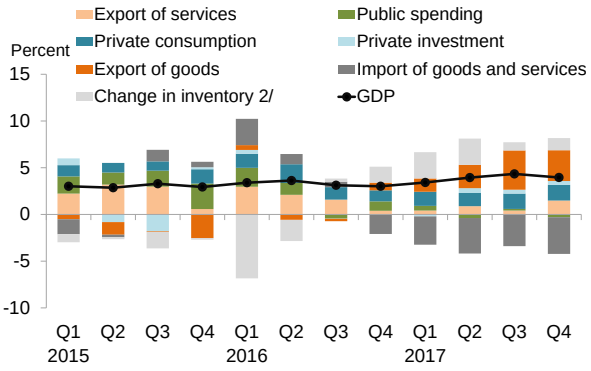


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

Thai economy

Economic growth continued to gain further traction, underpinned by strong merchandise exports and tourism growth as well as gradual improvements in private spending. Moreover, government measures aimed at improving welfare of low-income households were an additional growth driver. Meanwhile, public expenditure would remain an economic driver, despite more-than-expected impacts from the Public Procurement and Supplies Management Act.

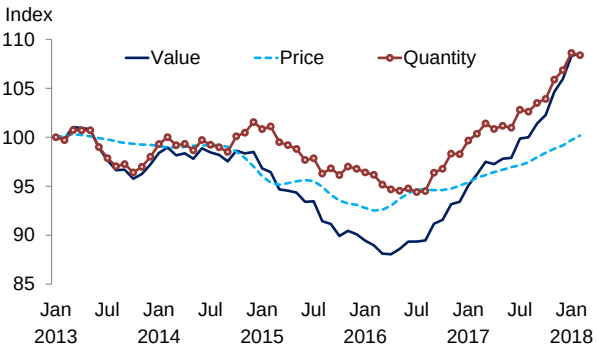
Contribution to Thailand's GDP growth[1/]



Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Board, calculation by Bank of Thailand

Thai exports (excluding gold): value, price, and quantity

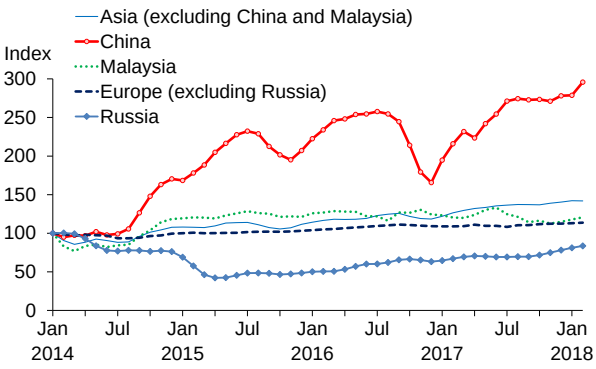
(3-month moving average, seasonally adjusted; January 2013 = 100)



Source: Customs Department and Ministry of Commerce, calculation by Bank of Thailand

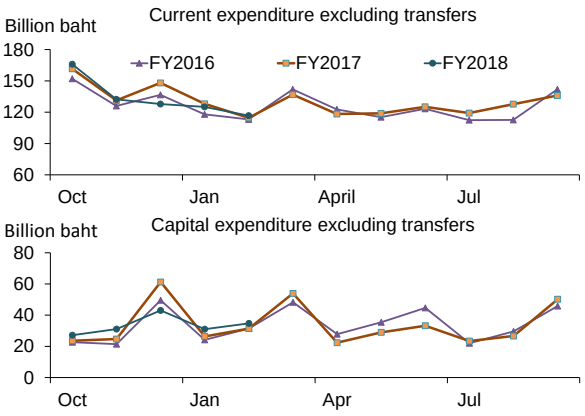
Index of foreign tourists classified by nationality

(3-month moving average, seasonally adjusted; January 2014 = 100)



Source: Department of Tourism

Public spending by central government

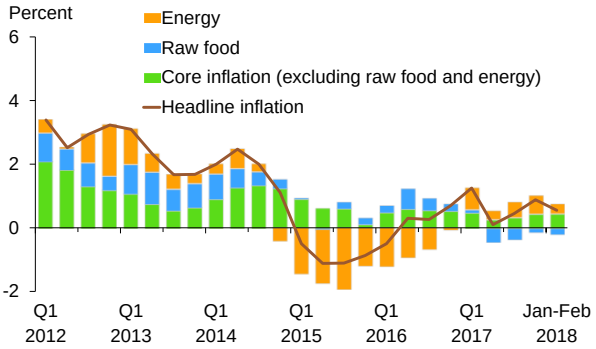


Sources: Bureau of Budget, Fiscal Policy Office

Inflation

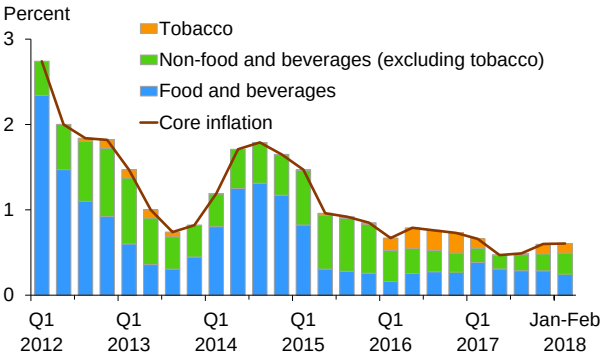
Headline inflation was expected to rise more slowly than the previous estimate due to fresh food prices. Meanwhile, core inflation remained at a low level similar to the previous assessment, with one-year-ahead inflation expectations of businesses and professional forecasters stabilized.

Contribution to headline inflation



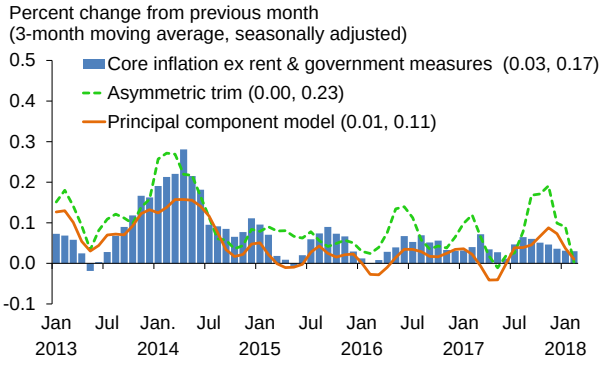
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculation by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculation by Bank of Thailand

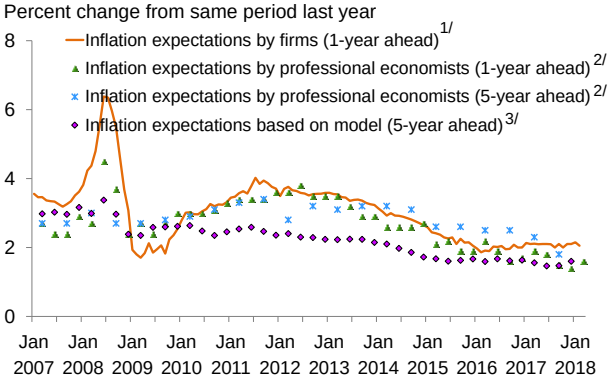
Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of February 2018, while the second value is 2004 - 2014 average; Asymmetric trim excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; Principal component model calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculation by Bank of Thailand

Inflation expectations

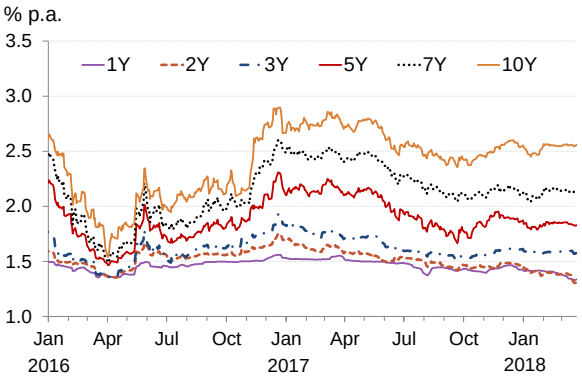


Sources: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial conditions

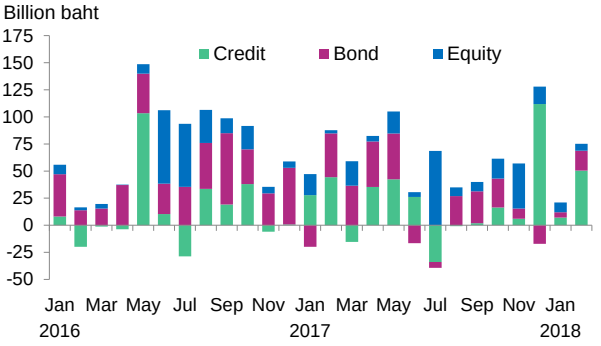
Short-term money market rates remained low. Nonetheless, short-term government bond yields were below the policy interest rate due to limited bond supply and increased demand from foreign investors. Long-term government bond yields gradually rose in line with U.S. Treasury yields. Total corporate financing continued to increase through various channels. The baht appreciated somewhat against the U.S. dollar, consistent with movements of most regional currencies due to U.S. dollar weakening. However, the baht appreciation was stronger relative to other regional currencies consistent with improving economic fundamentals.

Thai government bond yields



Source: Thai Bond Market Association (Thai BMA)

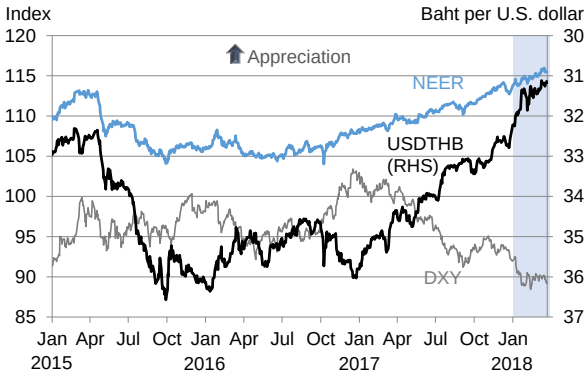
Total corporate financing by instrument*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.

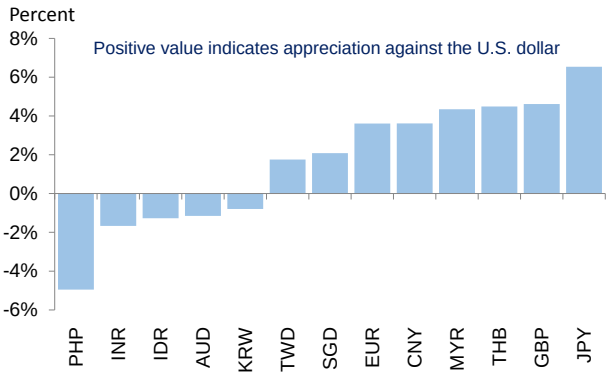
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and Dollar Index (DXY)



Sources: Bank of Thailand and Reuters (data as of 27 March 2018)

Currency movements vis-a-vis U.S. dollar (27 Mar 2018 compared with 29 Dec 2017)

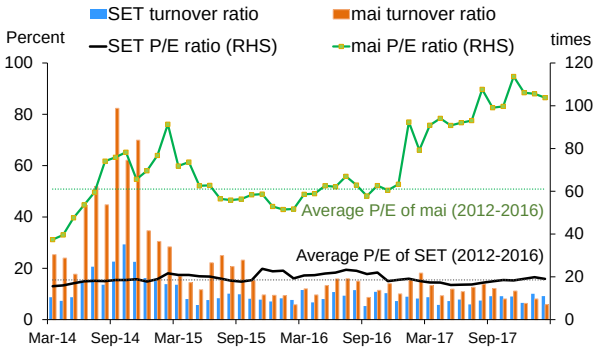


Sources: Bank of Thailand and Reuters (data as of 27 Mar 2018)

Stability: financial markets

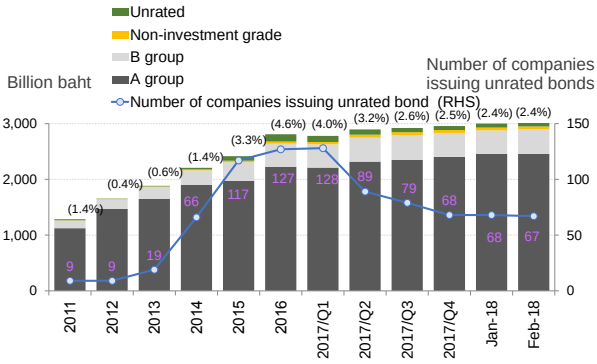
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand stayed close to the historical average. The P/E ratio of the Market for Alternative Investment (mai) remained high due to continued losses in agricultural and services sectors. New issuance of unrated bonds continued to slow down since the end of 2016 after some defaults.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of February 2018)

Corporate bonds outstanding



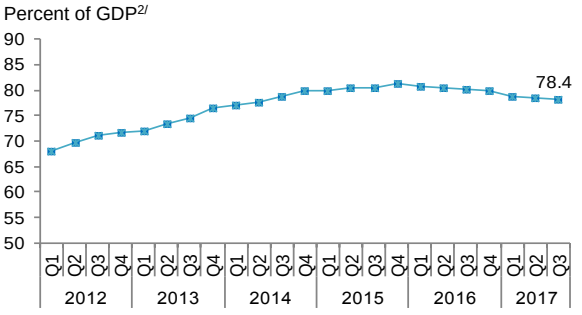
Note: ( ) represents percent of unrated bonds in total corporate bonds

Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

The ratio of household debt to GDP remained at a high level despite its continued declining. Deleveraging was mostly concentrated among high-income households. On the other hand, debt serviceability of households deteriorated as reflected in a higher NPL ratio in mortgage loans. This issue warranted monitoring going forward.

Household debt[1/]

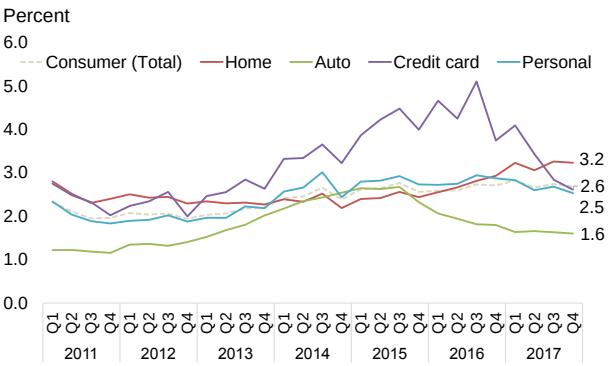


Note: [1/] Loans to households by financial institutions

[2/] Calculated by averaging the 4 latest quarterly GDP

Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

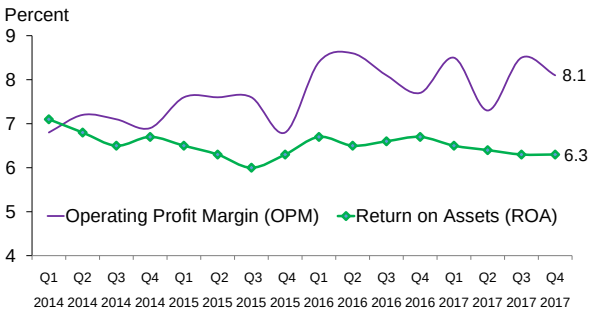


Source: Bank of Thailand

Stability: corporate sector

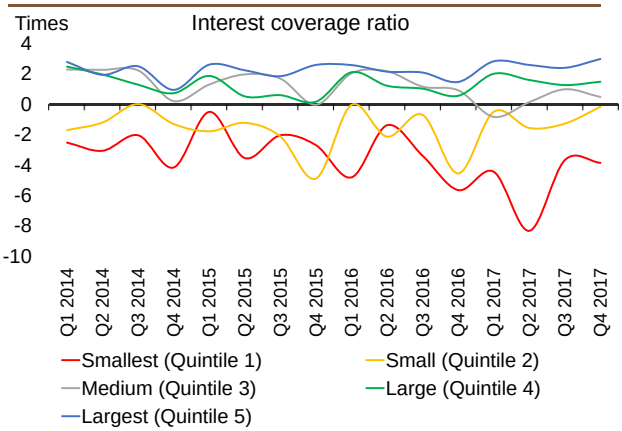
Overall stability in the corporate sector remained sound. Continued economic growth benefited financial positions of the corporate sector. As a result, profitability and debt serviceability of businesses improved. However, debt serviceability of small enterprises still deteriorated. Some businesses still incurred losses especially in manufacturing and services sectors.

Operating Profit Margin (OPM) and Return on Assets (ROA)*



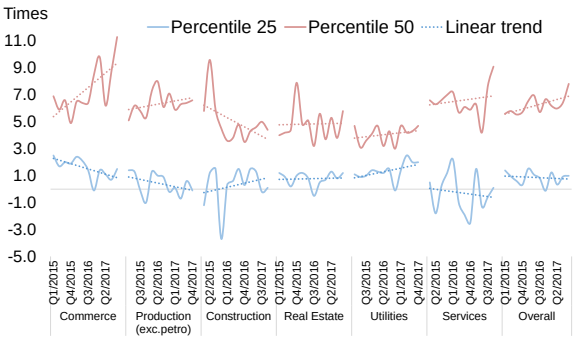
Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



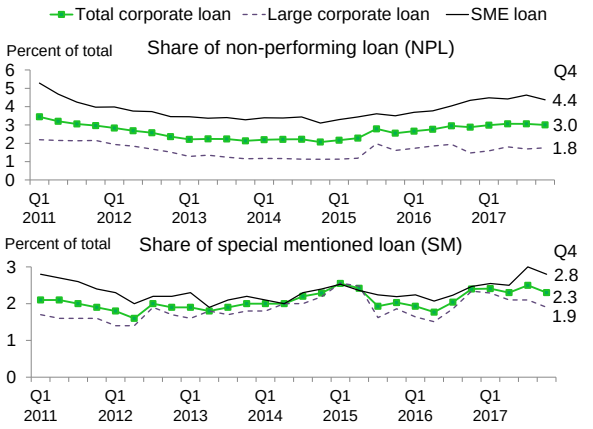
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Interest Coverage Ratio (ICR), classified by sectors



Note: * production exclude Petroleum and chemicals
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

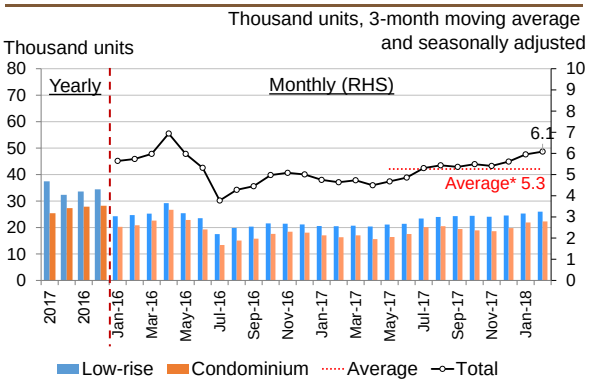


Source: Bank of Thailand

Stability: real estate

Demand for residences gradually expanded in line with economic conditions, as reflected in the number of residential units with newly approved loans that increased above the historical average. On the supply side, both newly opened low-rise and condominium residential units steadily declined since last year after the acceleration during the middle of last year. Meanwhile, prices of all real estate types trended upward.

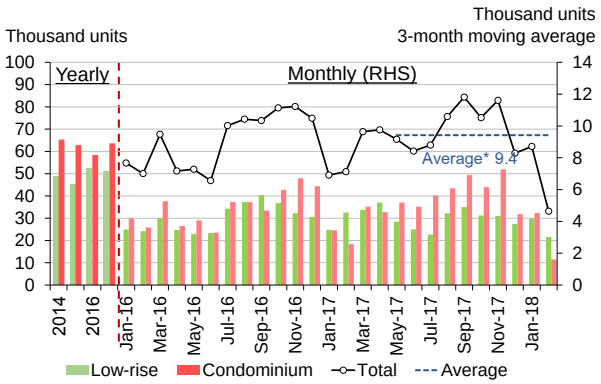
Residential units in Bangkok and vicinity with approved mortgages by commercial banks



Note: *Average during 2011-2017, excluding periods with government's stimulus measures (November 2015-April 2016)

Source: Bank of Thailand

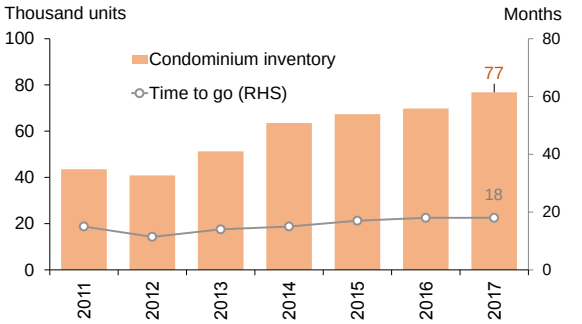
New residential projects launched in Bangkok and vicinity



Note: *Monthly average during 2011-2017, excluding the flood period (July - December 2011)

Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

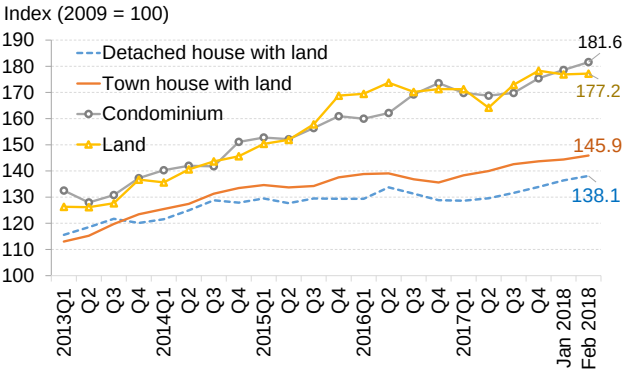
Condominium inventory in Bangkok and vicinity and 'time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold, using the average sales rate over the past 12 months.

Source: AREA and calculation by Bank of Thailand

Real estate prices

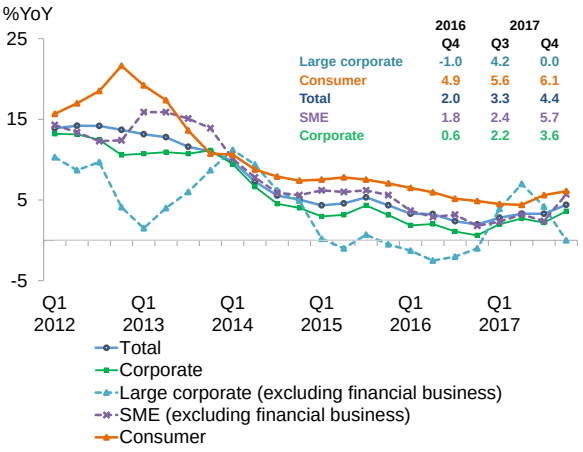


Source: Bank of Thailand

Stability: financial institutions

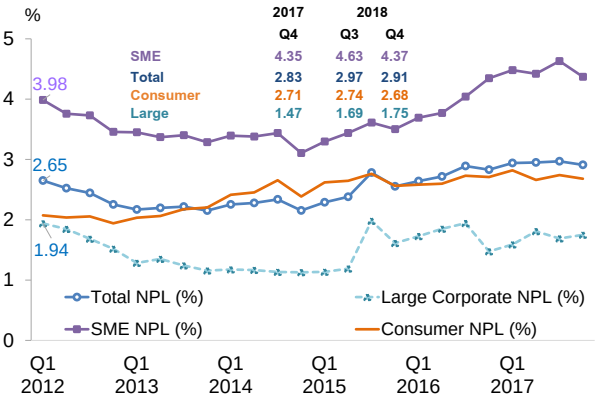
The financial system remained sound with high levels of provisions and capital buffers. In the fourth quarter of 2017, credit growth accelerated especially among business loans given to SMEs in trade sector and consumer loans for auto leasing purposes. Meanwhile, the NPL ratio slightly edged down, partly due to credit portfolio adjustments by financial institutions at the end of the year.

Credit growth in commercial bank system



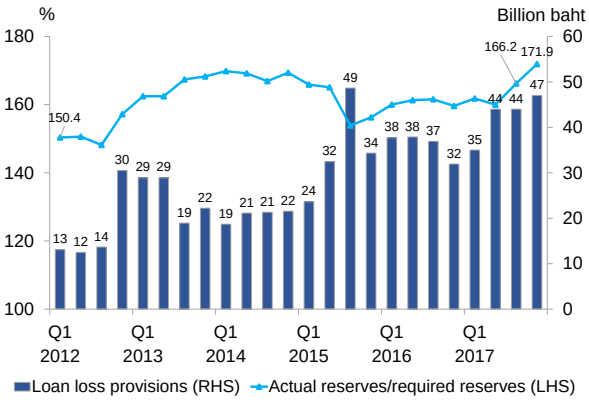
Source: Bank of Thailand

Non-performing loan (NPL)



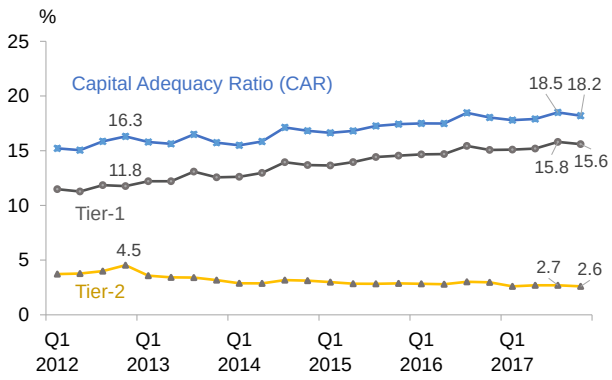
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

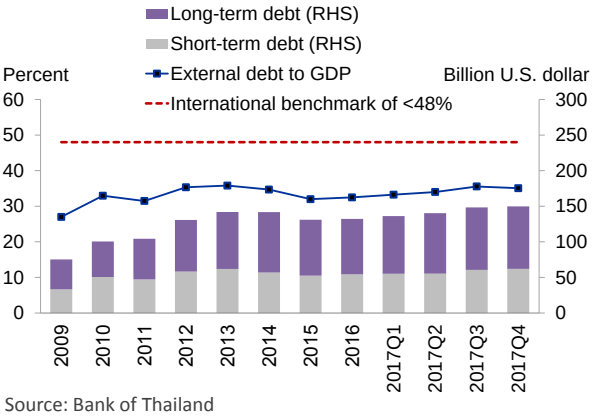


Source: Bank of Thailand

Stability: external position

Thailand's external stability remained strong due to a low level of external debt relative to an international benchmark with high level of international reserves relative to short-term debt.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt ratio

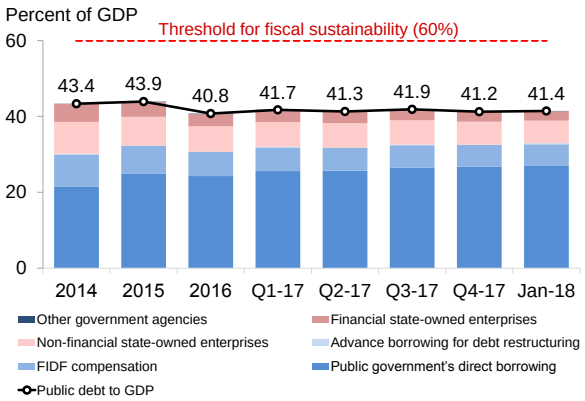


Source: Bank of Thailand

Stability: fiscal sector

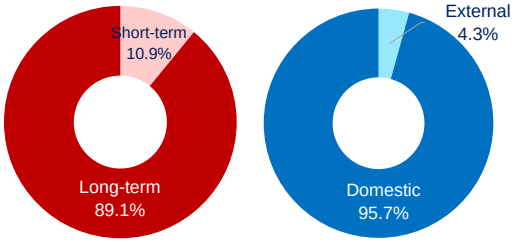
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the fiscal sustainability threshold.

Public debt to GDP ratio



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of January 2018



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office



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